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THESIS

**Forest change analysis in the Evros region using Landsat 8 satellite data and
Google Earth Engine Tools**

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ΕΘΝΙΚΟ ΚΑΙ ΚΑΠΟΔΙΣΤΡΙΑΚΟ ΠΑΝΕΠΙΣΤΗΜΙΟ ΑΘΗΝΩΝ

**ΣΧΟΛΗ ΘΕΤΙΚΩΝ ΕΠΙΣΤΗΜΩΝ
ΤΜΗΜΑ ΑΕΡΟΔΙΑΣΤΗΜΙΚΗΣ ΕΠΙΣΤΗΜΗΣ ΚΑΙ ΤΕΧΝΟΛΟΓΙΑΣ**

ΠΤΥΧΙΑΚΗ ΕΡΓΑΣΙΑ

**Ανάλυση δασικών αλλαγών στην περιοχή του Έβρου με χρήση δορυφορικών
δεδομένων Landsat 8 και εργαλεία Google Earth Engine**

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ABSTRACT

Protecting forested areas is very crucial for the environment and for the sustainability of every country. Especially in recent years where the rise of the global temperature has caused an increase in the frequency of forest wildfires. Crucial steps to combat that is the prevention, monitoring and the loss assessment of the damaged areas after the fact. The estimation of the loss is very important because it can play a crucial role in helping policy makers to take more targeted action and in general have more information to make better decisions for future planning. Satellite data can be of great help as a low-cost collection of data for the forested areas before, during and after the fact making it an important tool. Satellite imagery combined with remote sensing methodology, helps with the extraction of more information that can be extremely useful in situations like a forest fire either for a loss assessment or for the monitoring of the growth throughout the years. In this thesis Google Earth Engine, a free to use cloud-based geospatial analysis platform, alongside Google Collaboratory were used, Random Forest, Minimum Distance, and Mahalanobis classifiers were employed to analyze Landsat 8 data from 2013 to 2023 of the Evros regional unit and more specifically the protected forests based on the Ministry of Environment and Energy of Evros [\[5\]](#). The Random Forest classifier was the most accurate, achieving an overall accuracy of 0.988 and a 0.959 kappa coefficient. Significant forest losses were observed, in 2020-2021 due to small fires 411.052 km² were lost and 808.916 km² in 2022-2023 from a major fire event. This thesis showcases the usage of satellite data alongside cloud-based products, Google Earth Engine and Google Collaboratory, for forest loss assessment.

SUBJECT AREA: Remote Sensing

KEYWORDS: Remote Sensing, Forest Change Analysis

ΠΕΡΙΛΗΨΗ

Η προστασία των δασικών εκτάσεων είναι πολύ σημαντική για το περιβάλλον αλλά και για την βιωσιμότητα της κάθε χώρας. Ιδιαίτερως τα τελευταία χρόνια έχει υπάρξει δραματική αύξηση των δασικών πυρκαγιών λόγω της αύξησης της θερμοκρασίας της Γης . Σημαντικές ενεργείες για την καταπολέμηση των πυρκαγιών είναι η πρόληψη , παρακολούθηση και η εκτίμηση των καμένων εκτάσεων . Η εκτίμηση των χαμένων δασικών εκτάσεων είναι πολύ σημαντική διότι μπορεί να βοηθήσει τις υπεύθυνες αρχές να λάβουν πιο στοχευμένα μέτρα και γενικά να έχουν περισσότερες πληροφορίες για να λάβουν καλύτερες αποφάσεις για τον μελλοντικό σχεδιασμό βημάτων για την αναδάσωση των περιοχών αυτών. Τα δορυφορικά δεδομένα μπορούν να είναι πολύ χρήσιμα ως μία πλέων χαμηλού κόστους πηγή δεδομένων δασικών περιοχών πριν, κατά τη διάρκεια και μετά από οποιοδήποτε συμβάν. Τα δορυφορικά δεδομένα σε συνδυασμό με μεθόδους Τηλεπισκόπηση βοηθά στην εξαγωγή περισσότερων πληροφοριών που μπορούν να είναι εξαιρετικά χρήσιμες σε καταστάσεις όπως μια δασική πυρκαγιά, είτε για την αξιολόγηση της απώλειας δασικής έκτασης είτε για την παρακολούθηση της ανάπτυξης ενός δάσους κατά τη διάρκεια των ετών. Σε αυτή τη διπλωματική εργασία χρησιμοποιήθηκε το Google Earth Engine, μια δωρεάν πλατφόρμα γεωχωρικής ανάλυσης που βασίζεται στο cloud, μαζί με το Google Collaboratory. Οι ταξινομητές Random Forest, Minimum Distance και Mahalanobis χρησιμοποιήθηκαν για την ανάλυση δεδομένων Landsat 8 από το 2013 έως το 2023 της περιφερειακής ενότητας Έβρου και πιο συγκεκριμένα των προστατευόμενων δασών βασισμένα στο Υπουργείου Περιβάλλοντος και Ενέργειας του Έβρου [5]. Ο αταξινομητής Random Forest ήταν ο πιο ακριβής, επιτυγχάνοντας συνολική ακρίβεια 0.988 και συντελεστή kappa 0.959. Παρατηρήθηκαν σημαντικές απώλειες δασικών εκτάσεων, με 411.052 km² να χάνονται το 2020-2021 λόγω μικρών πυρκαγιών και 808.916 km² το 2022-2023 από ένα μεγάλο γεγονός πυρκαγιάς.

ΘΕΜΑΤΙΚΗ ΕΝΟΤΗΤΑ: Τηλεπισκόπηση

ΛΕΞΕΙΣ ΚΛΕΙΔΙΑ: Τηλεπισκόπηση, Ανάλυση αλλαγών δασικών εκτάσεων.

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This thesis represents the end of an era, as this part of my life comes to an end, I find myself both afraid and excited for the future. It was really challenging for me to find a subject that I am excited about, and this thesis really helped me organize my thoughts and plans for the future. This journey wouldn't be possible without the support, guidance, and encouragement of many individuals. I would like to express my gratitude to all of them. I am profoundly grateful to my advisor, Dr. Stavros Kolios, whose guidance and support were a very important element in the completion of this project. I would also like to give my gratitude to my parents who have been supporting me throughout my studies and all the friends who have been supporting me through the years aiding me through my personal growth.

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PREFACE

Forests are a crucial part of our planet and with the increase in temperatures and forest fires forested areas, a very important source of raw materials and protection from natural disasters, is getting lost making the economic and environmental state of the planet wors as times goes by. This thesis, focused on the forest change analysis of the Evros regional unit in Greece, was born out of my deep concern for environmental preservation and a desire to contribute to the research and showcase the use of cloud based tools in remote sensing [\[7\]](#).

The primary aim of this research is to assess the changes in forest cover in the protected areas of the Evros region over a decade, from 2013 to 2023. Utilizing cloud-based tools such as Google Earth Engine and Google Collaboratory, this study employs satellite data from Landsat 8 to analyze forest loss and regrowth. The insights garnered from this analysis are intended to help pinpoint areas that were burned during the decade, weather from the wild forest fires or just other environmental factors, and the areas that have gone through minimal regrowth.

The structure of this thesis is as follows: Chapter 1 introduces the study and its objectives. Chapter 2 reviews the methodology and background information that were required for this thesis. Chapter 3 presents the findings and analysis. Chapter 4 offers a discussion and conclusion, including recommendations for future research.

Undertaking this research has been a deeply enriching experience, allowing me to contribute to an area I am passionate about while enhancing my skills in geospatial analysis, environmental research alongside more technical skills like programing. It is my hope that this thesis can be an example of how free cloud-based tools can help in remote sensing projects with minimum equipment needed.

1. INTRODUCTION

1.1 Forests

1.1.1 About forests

Starting with the most important term for this thesis, the first question that should be answered is: what is a forest? According to Encyclopedia Britannica [\[1\]](#) “A forest is a complex ecological system in which trees are the dominant life-form.”. However, forests also foster a variety of animal species as well, including Mammals, amphibians, and insects alike together with different types of vegetation, including shrubs, grassy plants etc.

Forests are very important not only for the ecosystem that they support and are a part of, but also for humanity as a whole and more specifically the groups that live near them. Among many benefits they also help in various aspects of everyday life, they enhance the wellbeing of society, prevent flooding in the inhabited areas that they surround, keeping on equilibrium the human waste production like greenhouse gases and helping maintain good overall air quality in the pollution-dominant era that humanity is currently facing.

Along with the protection of humans against the air pollution which leads to several health issues associated directly with poor quality of atmospheric air, forests also contribute to a population's mental health by providing people with places to socialize and exercise helping maintain physical and psychological health by exercising in nature and in general providing a break in the intensity of everyday life, promoting healthier habits. Therefore, it is of great importance for today's societies to take care of their natural sites in order to upgrade the citizen's quality of life which subsequently leads to the general improvement - both in means of wealth and health.

There are many different types of forests around the world, each one of them has endured and developed through harsh changes on the Earth's surface cultivating an ecosystem unique for every location and the living beings that inhabit it. It is very important to determine ways to differentiate them in order to ease their study. In general forests can be categorized in three major branches: (1) Boreal forests, (2) Temperate forests and (3) Tropical forests [\[14\]](#).

- **Boreal forests** are usually found in cold permanently frosted areas. There are two different types of Boreal Forests. Open Canopy Boreal forests which can be found on higher latitudes and host small biodiversity. The second type is commonly known as Closed Canopy Boreal which as opposed to the former they can be found lower latitudes and provide a habitat for a wide variety of species,

- **Temperate forests** have four seasons, temperatures ranging from -30 to 30 °C and precipitation ranging between 75 cm to 1.5m per year. There are three types of temperate forests: Deciduous Forest, where their leaves fall in the winter and autumn months and the canopy is not dense, Coniferous Forest which can be found in cloudy areas where the climate remains cold for the entire year leaves do not fall during the deep wintertime. Last but not least, Temperate Rainforest is a type of Temperate Forest where the weather is average, but the precipitation levels can be extremely high over 3.5 m/year.
- **Tropical forests** can be found in regions where the climate is warm and between 20 and 30°C. In the broader view, the Tropical forests can be categorized into five different types. Starting with the Rainforests which are the most humid with a high level of precipitation, Moist forests where there is less rain and some seasonality, Dry forests where there is very little rain and the trees tend to lose their leaves in the dry season, Cloud forests, they tend to be in higher latitudes and are misty and dark and finally Mangroves which can be found on shores.

In Greece, the forest types that can be found are mainly Thermophilus deciduous forests, broadleaf evergreen forests, and coniferous Mediterranean forests [\[13\]](#). Understanding and preserving the delicate equilibrium of forests and actively attempting to coexist and protect these ecosystems serves the well-being of our planet and the prosperity of future generations. In this thesis the central theme consists of using remote sensing data and tools to analyze forest land changes using classification methods and tools.

In summary, forests are complex ecological “*havens*”, characterized by the dominant presence of trees that support a wide variety of life forms. Beyond their crucial role in sustaining biodiversity, these complicated ecosystems also serve as crucial sources of wellness for humanity, especially the populations that surround them. Forests also serve as a main defense system against floods, are well-structured and effective mitigators of greenhouse gases, and maintain air quality, thus protecting both the environment and public health. Every ecosystem has its own needs and when preserving and maintaining it scientists and relevant authorities should consider these needs. Monitoring these ecosystems using remote sensing methods is a crucial step to understand both the different forest ecosystems and how they grow overtime but also for the assessment of destroyed areas being able to procure that knowledge is very important because it can play a crucial role in managing forest after natural disasters and aid in reforestation efforts.

1.1.2 The Evros regional unit and its forest ecosystem.

The regional unit of Evros is part of the region of Thrace, the capital city of Evros is Alexandroupolis located southwest of the regional unit's coastline. Evros is a picturesque regional unit in northeastern Greece, known for its diverse landscapes, rich cultural heritage and the Evros Delta (the banks of the river Evros) wetlands that host a variety of migrating and local animal species.

The regional unit of Evros has two very important National Parks named Dadia-Lefkimi-Soufli and Evros Delta and also five Natura sites. These sites play a crucial role in fostering the biodiversity of the area, supporting the ecosystem, preserving the natural beauty of it and hosting 65 species of mammals, a wide diversity of reptiles and amphibians and 320 species of birds [3]. The main forest type in this region is Thermophilus deciduous forests [2], these types of woodlands form a broad yet inhomogeneous belt across the sub-Mediterranean zone of Europe. In Figures 1(a) and 1(b) two of the most important forests of the aerie are shown in maps by Natura [4][15]

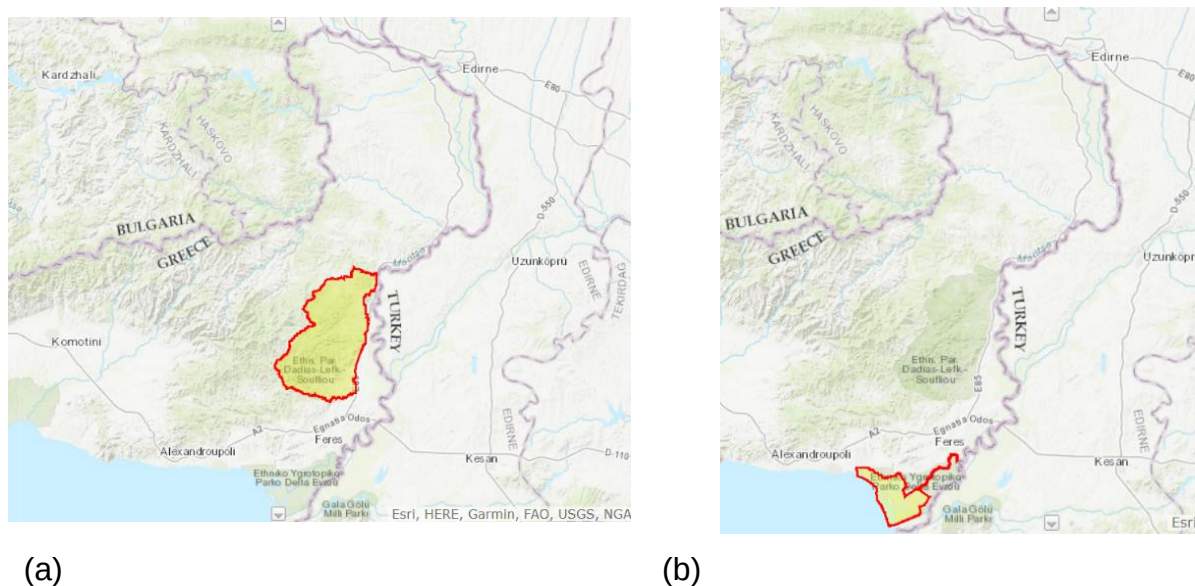


Image 1 Two of the most important Natura 2000 forests in the Evros region a) Dadia forest b) Delta Evrou

Inappropriate forestry interventions pose a significant threat, particularly in Central Europe over the years traditional management methods like coppicing, livestock grazing have been reduced making it a threat for the sustainability of the forested areas that have excelled due to these practices. Additional concerns include fires, overgrazing by game, and the spread of non-native species. Preserving the habitat requires nature-friendly forestry management or, in some instances, no intervention especially in the south or the restoration

of traditional management practices, particularly in the north [\[10\]](#). The region has over three hundred species of birds, more than fifty species of mammals and a variety of reptiles and amphibians. The region is also home to migrating birds in the eastern Mediterranean. It is clear that the surveillance and preservation of the forested areas of the Evros regional unit is crucial, especially in recent years. The entire region suffered a significant loss of 912.04 km² [\[22\]](#) due to a devastating wildfire during the summer of 2023. After a significant loss like this it is crucial to study the area in order to take the necessary steps to assist in the regeneration of the ecosystem to obtain its equilibrium again. Furthermore, analysis of the area can even help policy makers take action to prevent further destruction and choose ways to tackle any difficulties in the future based on data from previous years. By protecting the forested areas, the region's economic state will also get less affected in the long-term because of the loss of income due to tourism that stems from these natural sites.

1.2 Remote Sensing

Remote sensing is a pivotal process in acquiring information without direct physical contact and it is extensively employed across various domains such as agriculture, land use mapping, disaster management, climate monitoring, urban planning, weather forecasting, forest mapping, water management, mining, and more. This technique involves the utilization of sensors mounted on satellites, aircraft like drones, and meteorological balloons. The data gained from remote sensing gain even more meaning when they are used alongside ground-based information obtained from sources like research teams and ground stations, especially in remote or forested areas.

Satellite imagery stands out as a crucial source of remote sensing data for research initiatives, offering an extensive repository of spectral information and a vast historical record, enabling the tracking of Earth's evolution over time. Remote sensing facilitates researchers in gaining invaluable insights into trends and patterns through the application of tools and processes that give the spotlight to information that would otherwise be inaccessible.

In remote sensing, data collection relies on an array of sensors categorized into two main types: passive and active. Passive sensors, as their name suggests, do not necessitate energy to collect data, for example the human eye, which employs reflected light to construct an image. On the other hand, active sensors generate their own light pulses, and information

is derived from the time taken for the pulses to return to the sensor. This mechanism is similar to how bats navigate and perceive their surroundings.

Satellites equipped with remote sensing capabilities often incorporate one or multiple sensors tailored to capture specific spectral wavelengths known as bands. The selection of these bands is contingent upon the satellite's intended application. Standard bands include red, green, and blue, facilitating the creation of images in their "true" color, aligning with human perception of the visible light spectrum. Additionally, near-infrared wavelengths are frequently employed due to their ability to reveal unique information, such as burn marks on the Earth's surface.

In summary, remote sensing, facilitated by an array of sensors and sophisticated technologies, not only expands our observational capabilities but also contributes significantly to scientific research and decision-making processes across diverse fields. Helping scientists analyze and create solutions and quickly adapt to problems that would otherwise be very difficult to comprehend due to their large scale.

1.2.1 Data Levels

The data that are collected from satellites are categorized based on how processed the data are. There are four levels of satellite data products.

1 Level 0 Products

Level 0 products is a collection of raw data directly received from the satellite without any processing. Including telemetry information.

2 Level 1 Products

Level 1 products, when the raw data are processed to convert digital numbers into physical measurements like radiance or reflectance. Radiometric corrections are applied to account for sensor-specific distortions and atmospheric conditions.

3 Level 2 Products

Level 2 products involve further processing to correct geometric distortions caused by factors like sensor optics, platform motion, and Earth's curvature. Geometric corrections ensure accurate spatial representation of the imagery.

4 Level 3 Products

Level 3 products, additional processing for specific applications or analyses are applied in this product. This may involve atmospheric corrections, noise reduction,

image enhancement, and other specialized processing steps tailored to the needs of the product [\[6\]](#).

1.2.2 Image preprocessing

There are several preprocessing operations and the way they are implemented is dependent on the datasets, the analysis needs to be implemented based on the end product and the accuracy requirements of the end user. There are two main preprocessing steps, Radiometric calibration, and Correction of distortions geometric or atmospheric.

Radiometric calibration is a process that in the scope of satellite data happens two times in the launch, before, calibrating the satellite's sensor and after launch by calibrating the data by comparing ground truth datasets with the satellite data. Radiometric corrections are used to minimize the effects that are derived from atmosphere, viewing angle, and sensor characteristics. The most common issue in remote sensing are clouds, to solve that problem multiple images over several days can be taken, creating a composite image without clouds by taking the parts of the images that do not have clouds and clipping them together to an image composite.

Geometric correction is the process of adjusting images that might be distorted making the geographical representation of the image incorrect. This correction ensures that the spatial representation of the image is accurate to the real location that the image represents.

Atmospheric distortion is the distortion caused by the atmosphere, for example by aerosols like dust, water vapor or oxygen. These atmospheric characteristics impact radiance, causing scatter radiation and light absorption in different wavelengths based on the molecules that cause the disturbance. Atmospheric correction is very important to make sure that the true Earth's surface radiance is captured and not distorted by the disruptive atmospheric effects.[23]

1.3 Google Earth Engine

In recent years a lot of satellite data access portals have been developed making the accessibility of satellite images, for big time periods, possible for anyone who is interested in using them. This development in the scientific community alongside the creation of notebooks like Google Collaboratory or Jupyter Notebook and other online cloud-based

tools, have built the bases for the creation of Google Earth Engine. Google Earth Engine (GEE) is a cloud-based platform developed by Google that enables researchers, and developers to visualize and analyze large-scale geospatial data sets. It provides the user with an archive of satellite imagery and geospatial datasets with powerful cloud-computing capabilities making it a great tool for the monitoring and analysis of Earth's changes over time. It also provides its users with a library full of commonly used remote sensing tools and techniques like classification algorithms, data cleaning, accuracy assessment for classification algorithms etc. GEE is widely used in environmental monitoring and research supporting applications in various subjects including forestry, agriculture, water management, climate change, and disaster response. Researchers and policymakers use GEE to gain insights into complex environmental challenges. GEE also allows for researchers to work in two different environments, Google Earth Engine Code editor which provides the user with an online javascript environment where the GEE tools can be used but it also allows for the creation of assets like a table of polygons for sample classes or an image collection that can be shared with other users promoting cooperation, also the Google Earth Engine API, which gives the users a way to use GEE libraries and data via their own device or via Google Collaboratory, which is commonly used in machine learning, data analysis and education. Making Google Earth Engine tools a very user-friendly tool that allows users to take advantage of the abundance of satellite imagery available and the extensive background on remote sensing online allowing for new people to explore and learn more about remote sensing without an extensive background in programming and also allowing scientists to have a cloud-based option minimizing the need for personal powerful computing systems.

1.4 Landsat satellites

After the Apollo Moon-bound missions the idea of the Landsat program was born with the main goal being the creation of a satellite that is used for surveillance, research and the start of an ongoing record of earth observation data by placing the satellite in a polar orbit in order to have 14 full orbits per day with revisit time of 18 days. The first Landsat mission was launched on July 23, 1972, at an altitude of 900 km in a polar sun-synchronous orbit which served until January 6, 1978. There are four generations of Landsat missions, first generation being composed of Landsat 1, Landsat 2 and Landsat 3, second generation being Landsat 4 and 5 (who out leaved the expected lifespan) where the resolution had been increased, the revisit time became 16 days and the satellites made 14,5 orbits per day,

after that the 3rd generation of Landsat satellites was created with Landsat 6 (who was never in orbit) and Landsat 7 the first satellite to have only one sensor although in 2003 the scan-line corrector failed and after that the images provided by the satellite have noticeable striped with NAN data. The fourth generation of Landsat satellites consists of Landsat 8 and Landsat 9 who after all the errors and the knowledge accumulated from the previous generations serv today a massive network of countries who work together, conducting research and creating new solutions to problems. Having collected a more than 50 year record of earths and humanities history, Landsat has provided researchers all over the world with data to help make lives easier and create unique solutions for problems like forest fires, air pollution, human expansion, agriculture and so many more .

1.4.1 Landsat 8 satellite

The data that will be used for this thesis is raw data from Landsat 8 dataset. Landsat 8 was developed as a collaboration between NASA and the U.S. Geological Survey (USGS). Landsat 8 launched on February 11, 2013, from Vandenberg Air Force Base, California, on an Atlas-V 401 rocket, carries two scientific instruments, the Operational Land Imager (OLI) and the Thermal Infrared Sensor (TIRS). [\[16\]](#).

The Operational Land Imager (OLI), constructed by Ball Aerospace & Technologies Corporation, captures data in the visible, near infrared, and short-wave infrared segments of the spectrum. Its images have 15-meter panchromatic and 30-meter multi-spectral spatial resolutions, covering a broad 185 km (115 miles) swath [\[17\]](#).

The Thermal Infrared Sensor (TIRS) employs cutting-edge technology grounded in quantum physics to measure land surface temperature through its observation of two thermal bands. TIRS utilizes Quantum Well Infrared Photodetectors (QWIPs) to detect long wavelengths of light emitted by the Earth, specifically in the thermal infrared range, beyond human visual perception. These QWIPs present a cost-effective alternative to traditional infrared technology and were developed at NASA's Goddard Space Flight Center in Greenbelt, Md. The QWIPs employed by TIRS are sensitive to two distinct thermal infrared wavelength bands, enabling the separation of Earth's surface temperature from that of the atmosphere. The design operates on the intricate principles of quantum mechanics. Gallium arsenide semiconductor chips trap electrons in an energy state 'well' until elevated by thermal infrared light of a specific wavelength. This elevation produces an electrical signal, which is then read out and recorded to generate a digital image

As shown in Table 1 Landsat 8 has a spatial resolution mainly of 30 meters. There are nine (9) spectral bands including one (1) panchromatic band and two (2) spectral bands. Some of these bands will be chosen based on the indexes that will later be calculated in order to extract more information from the satellite images, this process is called feature extraction. [\[11\]](#)

Table 1 Wavelengths and names of the Bands of Landsat 8 satellite

Bands	Name	Resolution [m]	Wavelength [μm]
Band 1	Coastal Aerosols	30	0.43 - 0.45
Band 2	Blue	30	0.450 - 0.51
Band 3	Green	30	0.53 - 0.59
Band 4	Red	30	0.64 - 0.67
Band 5	Near-Infrared	30	0.85 - 0.88
Band 6	SWIR 1	30	1.57 - 1.65
Band 7	SWIR 2	30	2.11 - 2.29
Band 8	Panchromatic	15	0.50 - 0.68
Band 9	Cirrus	30	1.36 - 1.38
Band 10	TIRS 1	100	10.6 - 11.19
Band 11	TIRS 2	100	11.5 - 12.51

1.5 Feature Extraction

Feature extraction in remote sensing is a pivotal process that involves the identification and extraction of relevant information from remotely sensed data, enhancing our ability to interpret and analyze Earth's surface characteristics. This extraction is imperative due to the vast and complex datasets generated by sensors aboard satellites, drones, and other platforms. The objective is to extract meaningful features or patterns that contribute to specific applications, such as land cover classification, change detection, and environmental monitoring. Various techniques are employed for feature extraction, ranging from traditional methods to advanced machine learning algorithms.

Traditional methods include spectral indices, principal component analysis, and texture analysis, which emphasizes the statistical properties and spatial relationships within the data. On the other hand, machine learning algorithms, such as deep learning and

support vector machines, leverage the power of computational models to automatically learn and extract features, enabling more robust and accurate analysis. The choice of feature extraction method depends on the nature of the remote sensing data and the specific objectives of the analysis.

Successful feature extraction not only aids in pattern recognition but also plays a crucial role in improving the efficiency of downstream applications, contributing to informed decision-making in fields such as agriculture, forestry, urban planning, and disaster management. As technology continues to advance, feature extraction in remote sensing remains a dynamic and evolving field, constantly refining our ability to harness the wealth of information embedded in Earth observation helping scientists and policy makers create solutions to problems.

1.5.1 Calculating NDVI

NDVI stands for Normalized Difference Vegetation Index, and it is used to measure vegetation density and it is used as a metric for the health and density of vegetation. NDVI has been used since 1973, playing a crucial role in the monitoring of plant health and helping in applications of remote sensing in both forestry and farming. The NDVI is calculated using the reflectance values of the red and near-infrared bands of the electromagnetic spectrum. The formula for NDVI is shown in the equation 1.

$$NDVI = \frac{NIR - Red}{NIR + Red} [1]$$

where NIR is the pixel values from the infrared band. The output values of NDVI range from -1.0 to 1.0, with negative values mainly generated from clouds, water, and snow, and values near zero mainly generated from rock and bare soil [\[8\]](#).

1.5.2 Calculating BSI

The Bare Soil Index (BSI) is an index used in remote sensing to identify and map areas of bare soil, it is used as a metric for soil variation. BSI is commonly used alongside NDVI in mainly agricultural applications for soil mapping and crop identification. BSI is calculated using the reflectance values of the red and near-infrared bands of the electromagnetic spectrum. The formula for BSI is shown in the formula 2.

$$BSI = \frac{(Red + SWIR) - (NIR + Blue)}{(Red + SWIR) + (NIR + Blue)} \quad [2]$$

where NIR is the pixel values from the near infrared band and SWIR is the pixel values from the short-wave infrared band. The output values of BSI range from -1 to +1, with negative values mainly generated from clouds, water, and snow, and values near zero mainly generated from rock and bare soil [9].

1.6 Classification Processes

Data classification is the process of classifying data into categories based on certain characteristics. Classification is a method to organize data points and it is used from everyday life, for example classifying a piece of furniture based on the room that it is intended to be placed in, as well as in science fields like biology, machine learning, geoinformatics. The increasingly big demand for data processing from the start of the 20th century due to the abundance of data has, over the years, paved the way for the development of sophisticated algorithms for the automation of classification tasks in order to handle the increasing demand for data management.

There are two types of classifications, Unsupervised and Supervised. In Supervised classifications the user provides sample areas that are manually classified by the user. The sample area's values are then separated into training and test datasets and the classifier uses those to find a model that represents the dataset and classifies the pixels of the image. Unsupervised classifiers do not have a user input to set the classes so the classifier 'decides' the classes and classifies the image based on that. In this thesis supervised classification methods will be used in order to have control over the classes and their overall classification process. Data classification helps in identifying sensitive information, reducing storage costs, and streamlining data management processes.

1.6.1 Minimum Distance classification process

The use of minimum distance classification methods in remote sensing is very commonly used. The minimum distance classification is an unsupervised classification used in pattern recognition and machine learning algorithms for classifying objects in certain user defined categories.

The basic logic behind a minimum distance classifier is to assign new data points to the class that has the minimum distance from that data point's value. How this classification

method works is, first of all a centroid is calculated which typically is the mean or average of the feature vectors of all the training data provided by the user. In the minimum distance classification, the distance between the new data point and the classes' centroid is calculated using the Euclidean distance. The Euclidean distance is calculated based on the formula 3.

$$D(A,B) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} \quad [3]$$

Here A (x1, y1) and B (x2, y2) are the coordinates, and their distance is calculated using the Pythagorean theorem.

After that Centroid is calculated for every class and now Each centroid will be used to calculate the distance between new data points to be classified. For example, If D1 The distance between a new data point and the centroid of Class 1 then to the distance between a new data point and the centroid of Class 2, the new data point will be classified as Class 1, if $D_1 < D_2$ and If the opposite is true. ($D_2 < D_1$) Then the new pixel will be classified as Class 2.

The main advantage of a minimum distance classification is that it is easy to understand and implement because the idea of assigning a point to the class with the nearest prototype aligns with our intuitive understanding of similarity. The main disadvantages are that since it relies on distances, outliers can significantly influence the classification and that Euclidean distance may not be appropriate if class distributions have different variances.

While minimum distance classification is straightforward, it may not perform well in complex scenarios with overlapping classes or varying class distributions. More sophisticated classification algorithms, such as support vector machines or neural networks, are often preferred for more challenging tasks. [\[24\]](#)

1.6.2 Random forest classification process

Similar to minimum distance classification, the application of Random Forest classification methods in remote sensing is widespread, offering a supervised approach within pattern recognition and machine learning algorithms. This method facilitates the classification of objects into predefined categories specified by the user.

Random Forest operates by utilizing a number of decision trees. Each tree is trained on a random subset of the training data and features, contributing to a collective decision-

making process. The algorithm employs a voting mechanism, where each decision tree independently predicts the class for a new data point, and the final classification is determined by the majority vote.

The strength of Random Forest lies in its ability to handle complex relationships within the data, leading to high accuracy and robust performance. Unlike the simplicity of minimum distance classification, Random Forest's ensemble nature helps mitigate overfitting issues, making it resilient to noisy features in the training data.

Additionally, while Random Forest offers notable advantages, it comes with complexities and computational demands. The algorithm operates as a "black box," making it challenging to interpret individual decision trees. An important thing to note is that the algorithm's inner workings are not as straightforward as the intuitive logic behind minimum distance classification and because of that bigger computational power is necessary.

In summary, Random Forest classification proves to be a powerful and accurate tool for remote sensing tasks, excelling in scenarios with intricate data relationships. Its ability to handle complex datasets and provide insights into feature importance makes it a preferred choice for various remote sensing applications [\[25\]](#).

1.6.3 Mahalanobis classification process

The Mahalanobis classification method is an approach frequently used in remote sensing applications. Unlike the Euclidean distance used in minimum distance classification, Mahalanobis distance is used for the Mahalanobis classification there main difference being that Mahalanobis distance accounts for how much the variables affect one another, making it a more robust measure.

In Mahalanobis distance classification, the first step involves calculating the covariance matrix from the training data for each class. This matrix represents the relationships and variabilities among different features, offering a more accurate measure of distance. The Mahalanobis distance (DM) between a new data point and the centroid of a class is then computed using the formula 4 shown below.

$$DM = \sqrt{(x - \mu)^T \Sigma^{-1} (x - \mu)} \quad [4]$$

Here, x is the vector of feature values for the new data point, μ is the centroid vector for the class, and Σ^{-1} is the inverse of the covariance matrix.

The classification decision is made by comparing the Mahalanobis distances for a new data point to the centroids of different classes. The data point is assigned to the class with the smallest Mahalanobis distance, reflecting a more nuanced understanding of the data's distribution. One notable advantage of Mahalanobis distance classification is its ability to account for correlations and variations in scale, making it less sensitive to outliers compared to the Euclidean distance. However, similar to other methods, it may still face challenges in scenarios with complex class distributions or overlapping classes.

In summary, Mahalanobis distance classification stands out as a sophisticated approach in remote sensing, offering improved accuracy by including the relationship between the variable and providing a more nuanced assessment of distances [25].

1.7 Accuracy assessment

1.7.1 Introduction

Accuracy assessment is a crucial step in evaluating the quality of classification results derived from remotely sensed data. It involves comparing the classified image to ground truth data to quantify the difference between the estimated classes and the actual land cover. This process is essential for determining the accuracy and reliability of the classified maps, which is vital for various applications such as land cover mapping and environmental monitoring. The assessment can be a good measure of quality involving the use of specific error metrics and techniques to measure the accuracy of the classification results such as calculating the confusion matrix and its derivatives like F1 score specificity and sensitivity as well as overall accuracy and Kappa coefficient [20].

1.7.2 Confusion matrix

A confusion matrix is a fundamental tool in the field of machine learning and statistics that is used to assess the performance of a classification model. It presents a clear and concise summary of the model's predictions by comparing them to the actual outcomes. The matrix is a table with rows and columns representing the predicted and actual classes, respectively as shown in an example table for two classes Class A and Class B.

Table 2 : Confusion matrix explanation

	Positive	Negative
Positive	TP	FP
Negative	FN	TN
	Class A	Class B

It consists of four main components: true positives (correctly predicted positive instances), true negatives (correctly predicted negative instances), false positives (incorrectly predicted positive instances), and false negatives (incorrectly predicted negative instances). These elements help in evaluating key performance metrics such as accuracy, precision, recall, and F1 score. The confusion matrix provides valuable insights into the strengths and weaknesses of a model, aiding in the refinement and optimization of predictive algorithms [\[23\]](#) [\[20\]](#).

1.7.3 Accuracy

Accuracy is used as a measure for how correctly the data points of an image are classified and it is calculated by using the elements of the confusion matrix (True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN)) [\[20\]](#) The accuracy is calculated based on the formula 5 below.

$$Accuracy = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}} = \frac{TP+TN}{TP+TN+FP+FN} \quad [5]$$

It is important to note that accuracy is a grate measure for a general idea of how well a model is performing but in cases where the classes are imbalanced other metrics like recall, precision or F1 score tend to be better tools to indicate the model's performance.

1.7.4 F1 score

The F1 score is a metric commonly used in machine learning and statistics to assess the performance of a classification model. It is particularly useful when dealing with imbalanced datasets, where one class significantly outnumbers the other. The F1 score is the harmonic mean of precision and recall, combining both metrics into a single value. Precision measures the accuracy of positive predictions, while recall, also known as sensitivity or true positive rate, quantifies the model's ability to capture all positive instances. The F1 score is calculated using the formula 6 shown bellow.

$$F1 \text{ score for class } x = \frac{TP}{TP + 0.5 * (FP + FN)} [6]$$

Where TP are the True Positives of class x, FP are the False Positives of class x, FN are the False Negatives of class x.

F1 score ranges from 0 to 1, with higher values indicating better model performance. A balanced F1 score suggests that the model maintains a good trade-off between precision and recall, making it a valuable metric for evaluating classifiers, especially in situations where the cost of false positives and false negatives is not the same [\[26\]](#).

1.7.5 Specificity

Specificity measures the ability of a model to correctly identify negative instances out of all actual negative instances. It is calculated as the ratio of true negatives to the sum of true negatives and false positives. The specificity is calculated based on the formula 7 bellow:

$$Specificity \text{ for class } X = \frac{TN}{TN + FP} [7]$$

Where TN are the True Negatives of class x and FP are the False Positives of class x .

Specificity is important when the cost of false positives (incorrectly identifying a negative instance as positive) is high, as it focuses on the model's ability to avoid false alarms [26].

1.7.6 Sensitivity

Sensitivity (True Positive Rate, Recall): Sensitivity measures the ability of a model to correctly identify positive instances out of all actual positive instances. It is calculated as the ratio of true positives to the sum of true positives and false negatives. The formula for sensitivity is shown in the formula 8 bellow.

$$\text{Sensitivity for class } X = \frac{TP}{TP+FN} [8]$$

Where TP are the True Positives of class x, FP are the False Positives of class x ,FN are the False Negatives of class x .

Sensitivity is crucial when the cost of false negatives (missed positive instances) is high, as it emphasizes the model's ability to capture all relevant positive cases [26].

1.7.7 Producer's accuracy & User's Accuracy

Producer's accuracy is used to show the classification's accuracy from the point of view of the producer (creator). More specifically it is used to measure the frequency of the correctly classified features on a classified image. It is calculated by dividing the number of correctly classified data points and the total number of training data for the specific class. On the other hand, the User's accuracy is used to measure the images accuracy in reference to the user not the maker. Indicating the probability of a pixel of a specific class to be actually on the ground. It is determined by dividing the total number of correct classifications for a specific class by the row total for that class [26].

1.8 Research objective

The primary objective of this thesis is to conduct an in-depth examination of the dynamic changes happening in the forest land cover within the Evros regional unit over a period of ten (10) years, with a specific focus on the summer months. The methodology employed in this investigation revolves around the utilization of state-of-the-art remote sensing techniques. The key aim is to precisely quantify and analyze the changes in land cover within the forested area, aligning with the areas mentioned in the guidelines set by the National Ministry of the Environment and Energy for the former Evro region [3].

This research holds significant importance as it serves as a foundation for future work in the matter and demonstrates the usage of cloud-based tools to analyze a forested area

in the span of ten (10) years. The insights derived from this thesis can play a pivotal role in facilitating different ways to approach a remote sensing forest change analysis.

Furthermore, due to the recent forest fires not only in the regional unit of Evros but also in the whole country of Greece it is imperative [\[7\]](#) that analysis like this happen because they can help, relevant authorities, policy makers and other sources to take more informed action and plan accordingly based on the information that can be acquired from studies like this. The forested areas not only serve as a buffer against environmental adversities but also provide crucial habitats for diverse local species. The observed alterations in land cover, if left unaddressed, pose a direct threat to the delicate balance of these ecosystems, leading to habitat loss. This, in turn, could have cascading effects on the populations of various local species, necessitating a multifaceted approach to conservation and ecological restoration.

In summary, this thesis endeavors to contribute not only to the scientific understanding of land cover changes in the Evros regional unit using relatively new tools but also to provide actionable insights for policymakers and environmental stewards. By anticipating, mitigating the potential risks associated with aerial destruction and habitat loss, the research aims to pave the way for sustainable management practices that will ensure the long-term resilience and vitality of the Evros region's ecosystems.

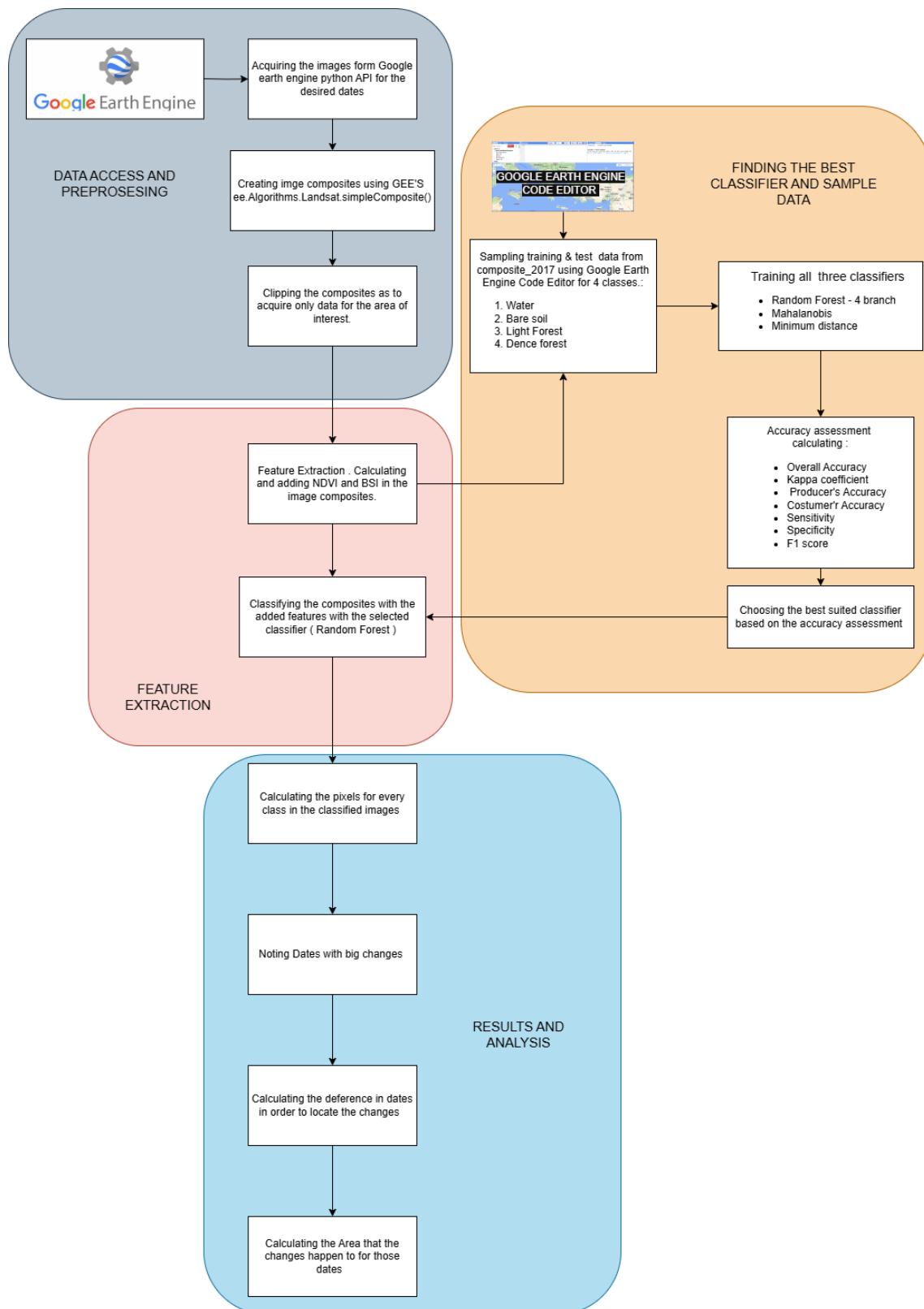
2. METHODOLOGY

2.1 Introduction

The process that was applied can be simplified in four steps, data access and preprocessing where the satellite images were acquired using the Google Earth Engine API [\[12\]](#), creating image composites for the summer months from 2013 to 2023 and clipping to the shapefiles based on the Ministry of Environment and Energy, clipping the forested areas, finding the best classifier and sample data for classification, using Google Earth Engine Code Editor to make an asset with the sample data for the training of the classifiers, then to make sure the best classifier for this dataset was used three classifiers were implemented and then an accuracy assessment was implemented for the classifiers, the best one was chosen, feature extraction calculating indexes in the image composites, applying the classifier that was chosen in the previous step, finally results and analysis by calculating the pixels for every class for all the classified images in order to note the differences in the time series

,after that locating the changes on the area and calculating the forest loss and gain in those years where the changes happened ,this process is shown in more detail in Diagram 1.

Diagram 1 The steps for the process of this analysis



2.2 Data access and preprocessing

2.2.1 Region Of interest

The Evros region is located in northeastern Greece, sharing borders with Turkey to the east and Bulgaria to the north. It is named after the Evros River, which flows through the region. The regional capital is Alexandroupoli, a city situated on the coast of the Aegean Sea. The Evros region is part of the Eastern Macedonia and Thrace administrative region in Greece as shown in the **image 2**.

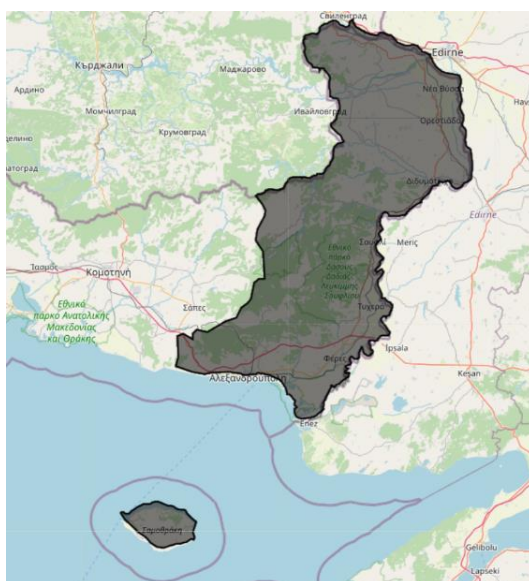
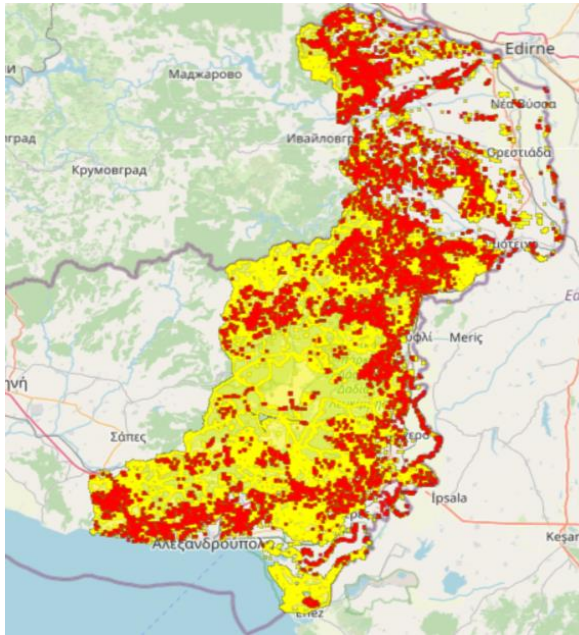
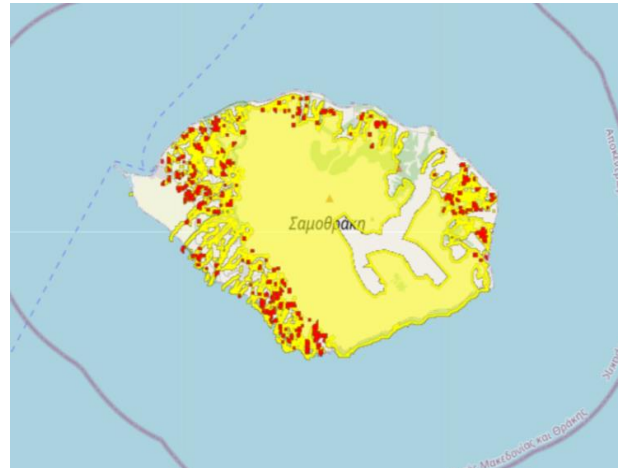


Image 2 : The regional Unit of Evros

For this research the main focus are the forested areas of Evros based on the maps from the dataset [5] which is a collection of the forest maps of the region documenting areas and characterizing them based on the forest or grassland features. These areas are protected by law and publicly available maps have been produced. The purpose of these maps is to define the areas in order to protect the ownership status from individuals protecting forested areas from overgrazing or illegal deforestation. For this analysis the mapped areas that have specifically two codes will be used, the land with code's 'DA' which means forest land that is cleared and code 'DD' meaning forest area that has consistently stayed forested shown in **Images 3 and 4**.



a)



b)

Image 3 Ministry of Environment and Energy shapefile, red: forest land that is cleared and yellow forest land that has consistently stayed forested a) main land b) island of Samothraki

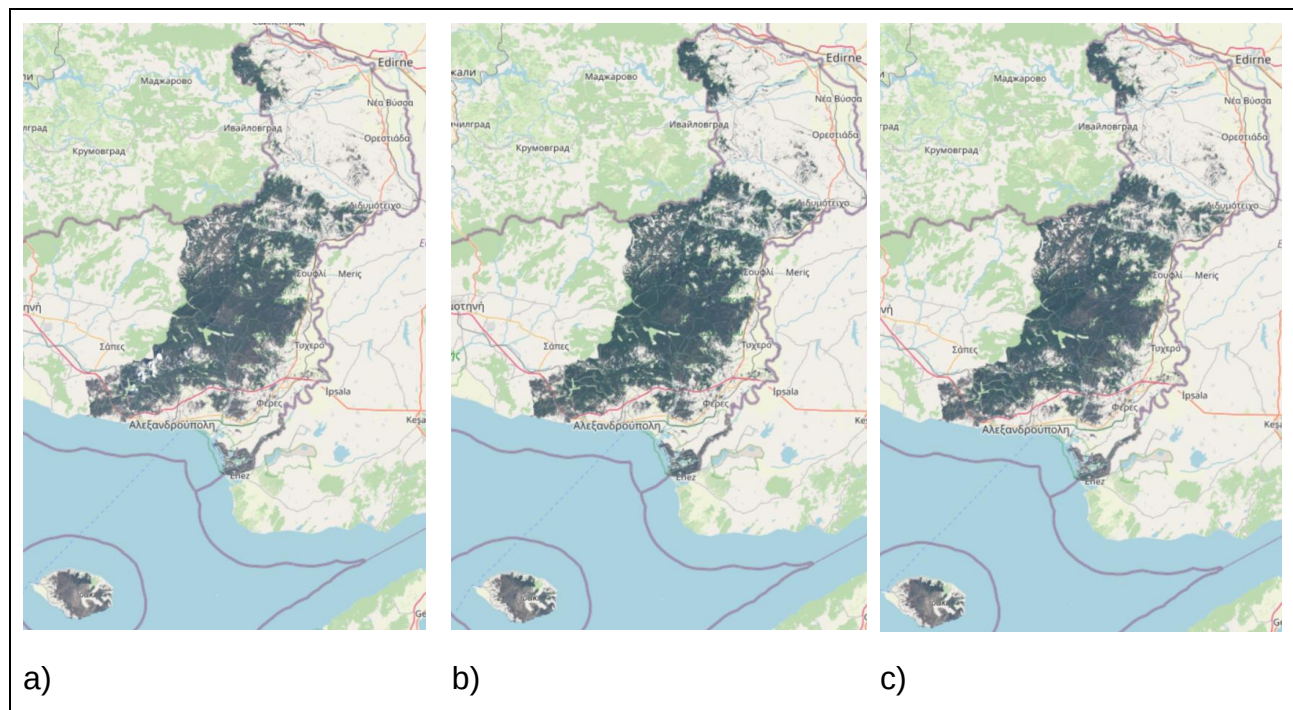
2.2.2 Data Collection

The data that will be used for this thesis are raw data from the Landsat 8 dataset from Google Earth Engine's data catalogue. For the purpose of this analysis the bands that will be used are the RGB (Red Green Blue), Landsat 8 bands B4, B3, B2 as well as NIR (Near Infrared) and the first SWIR (Short Wave Infrared), Landsat 8 bands B5, B6 as shown in **Table 3**.

Table 3 Bands of Landsat 8 that were used for this analysis and their Bandwidth.

<i>Bands</i>	<i>Names</i>	<i>μm</i>
Band 2	Blue [30 m]	0.450 - 0.51
Band 3	Green [30 m]	0.53 - 0.59
Band 4	Red [30 m]	0.64 - 0.67
Band 5	Near-Infrared [30 m]	0.85 - 0.88
Band 6	SWIR 1 [30 m]	1.57 - 1.65

The initial dataset for this analysis is a collection of images of the area of interest in the summer months from 2013 to 2023. In order to have clean images with very low cloud coverage the simple-composite tool for Landsat images from the gee (google earth engine) python API was used. The months that were chosen based on how clear the images are and the final image composites that were selected for every year from 2013 to 2023 are shown in **Table 4 (a to k)**.

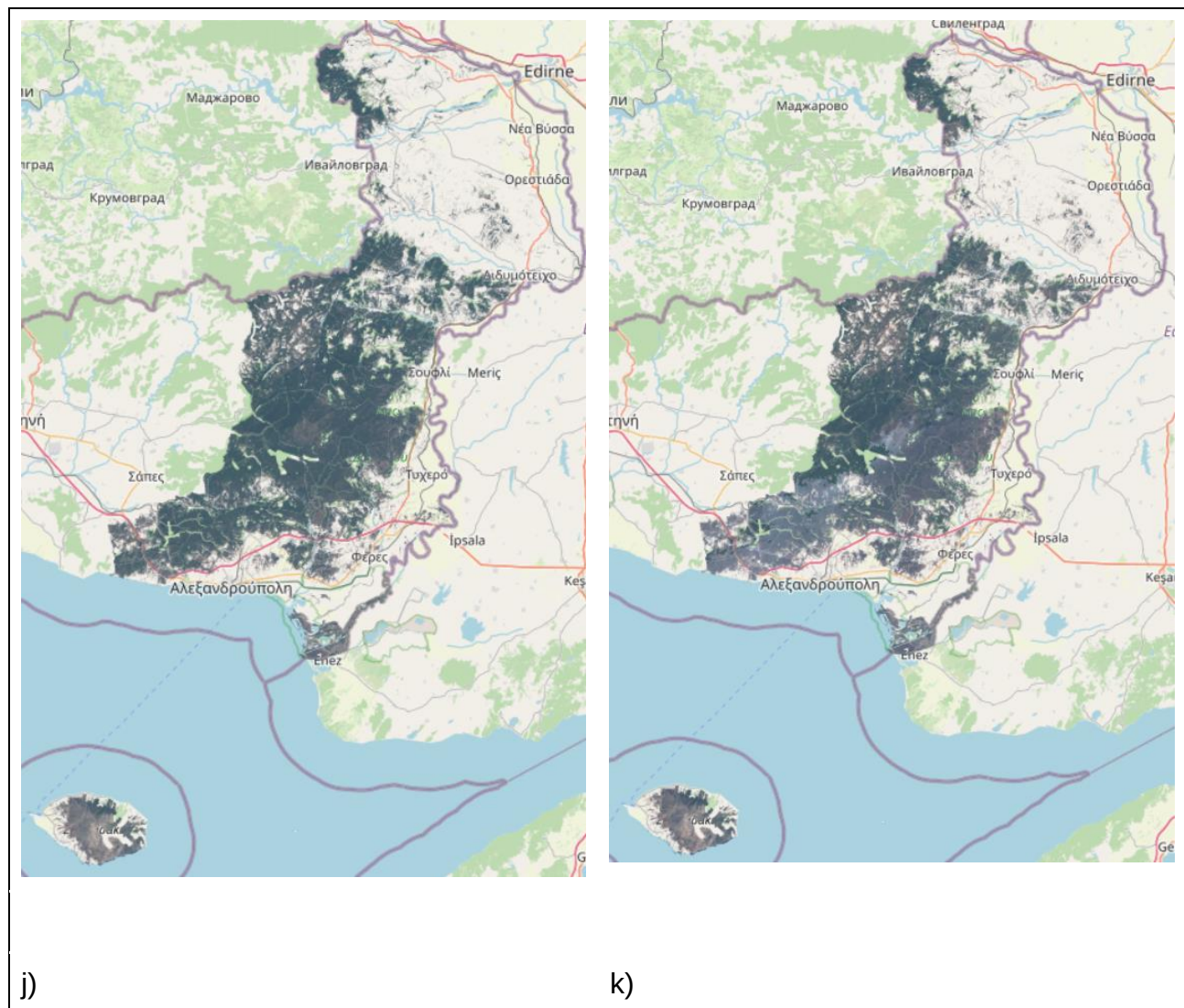
Table 4 Selected clipped images from 2013 to 2023.



f)



i)



2.3 Finding the best classifier and sample data for classification

2.3.1 Classification Algorithm Training

The classifiers that were implemented in the process for this project were Random Forest with four branches, Minimum distance and Mahalanobis classifiers using the google earth engine python api. This was done in order to choose the one that feels most the analysis of our data. To determine that an accuracy assessment was made for all of the classifiers and then they were compared with each other to see which one would give the best possible classification given the training dataset that were previously made.

2.3.2 Accuracy Assessment

Accuracy assessments are crucial in evaluating the performance of classifiers and making informed decisions about their effectiveness. In this process, the first step involves acquiring the confusion matrix using the GEE API's tools for each classifier under consideration. Subsequently, key metrics such as specificity, sensitivity, and F1 score are calculated based on this confusion matrix, as highlighted in paragraph 1.7.2. These metrics provide a comprehensive understanding of a classifier's ability to correctly classify instances, especially in the context of true positives, true negatives, false positives, and false negatives. The purpose of conducting accuracy assessments is to enable a thorough and meaningful comparison of different classifiers. Using these metrics, and practitioners, valuable insights can be acquired into the strengths and weaknesses of each classifier, aiding in the selection of the most suitable model for a given task.

2.4 Feature Extraction

The indexes that were added to the composites are NDVI and BSI using the aforementioned formulas (see chapter 1.3) which help with the distinction of forest land and the bare soil in the classification process. The classes for the classification process are Water, Light Forest, Dense Forest, and Bare Soil. The samples were extracted for the image composite from 2017 mentioned in the previous paragraph using the Google Earth Code Editor creating an gee table asset with the polygons of the samples.

The selection of polygons in this study is a meticulous process because in order to have the best outcome there is a lot of trial and error when choosing the sample polygons in order to have the best representation of the different characteristics for each class. By choosing polygons based on the similarity of values, it ensures that the selected samples provide a balanced representation of each class. This approach is important in order to minimize potential biases that might emerge if the chosen polygons do not represent correctly the variability within the dataset. The emphasis on similarity fosters a more nuanced understanding of the classes under examination, ultimately contributing to the reliability of the analysis. It allows for a more comprehensive evaluation of the data, enabling the extraction of meaningful conclusions helping to make informed decisions based on a representative selection of polygons. A main note for the classes of this classification is that the high similarity of bare land and Dense Forest with the Light forest class is to be expected

because the light forest class is a mix between the two. The similarity that was calculated can between the samples of every class be seen in **Table 5**.

Table 5 Similarity between selected classes based on sample data.

class A	class B	similarity
Water	Bare soil	0.3%
Water	Dense Forest	3.2%
Bare soil	Dense Forest	11.2%
Light Forest	Dense Forest	75.0%
Light Forest	Water	2.5%
Light Forest	Bare soil	3.7%

3. RESULTS AND ANALYSIS

3.1 Overview of Findings

3.1.1 Performance of Classifiers

For each classifier mentioned previously, Random Forest, Minimum Distance, and Mahalanobis, confusion matrices and their derivative metrics were calculated using the GEE (Google Earth Engine) Python API. These metrics included Sensitivity, which determines the ability of a classification method to accurately make distinctions between various classes or features in the imagery, Specificity, a measure of the algorithm's precision in identifying pixels or areas that do not belong to a specific class, F1 Score, Producer's Accuracy, and Consumer's Accuracy. These metrics were calculated individually for each class, namely Water, Bare Soil, Light Forest, and Dense Forest, as detailed in the respective tables (**Tables 6,7,8**) for the classified image composite of 2017.

In addition to these metrics, overall accuracy and the kappa coefficient were also determined for each classifier, offering a complete evaluation. The kappa coefficient serves as a way to compare classification results against randomly classified values. The results for all classifiers are summarized in the provided **Table 9**.

Diagram 2 Confusion matrix for Random Forest classifier with 4 branches (tested on 2017 image).

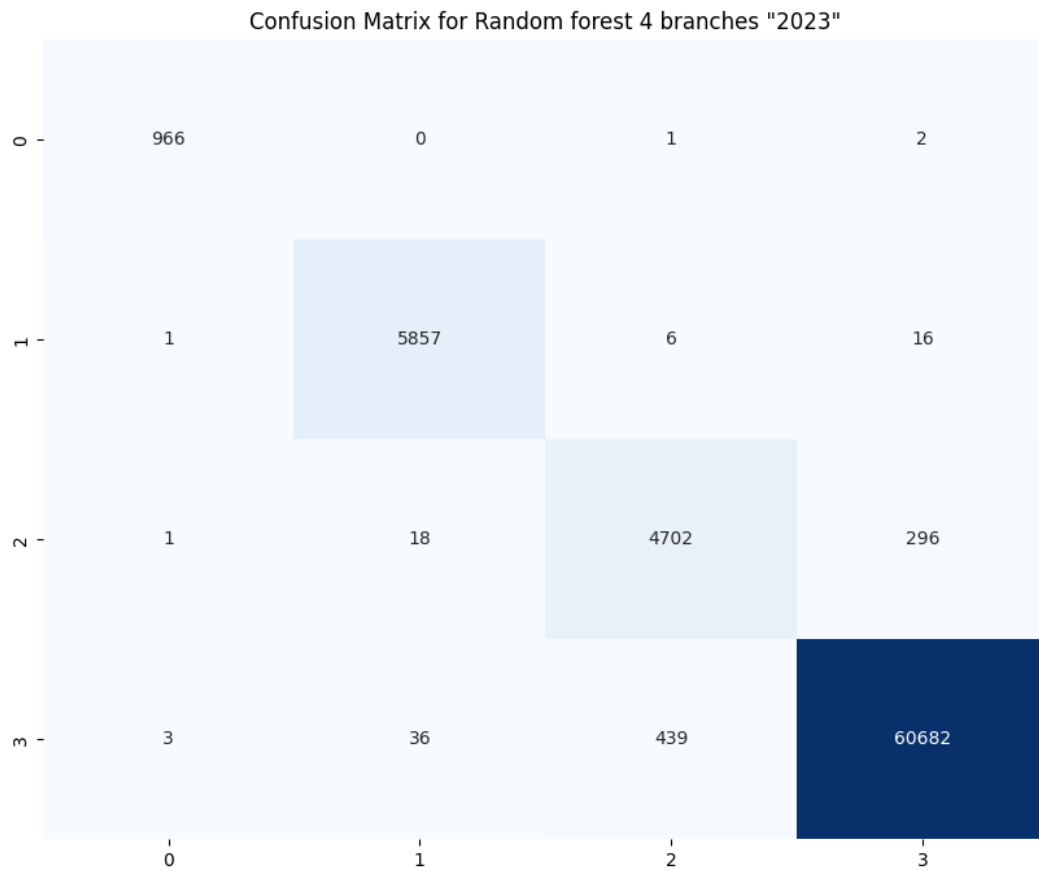


Table 6 Producer's Accuracy Consumer's Accuracy, Sensitivity, Specificity and F1 score for Random forest Classifier.

Class	Producer's Accuracy	Consumer's Accuracy	Sensitivity	Specificity	F1 score
0 [Water]	0.996	0.995	0.996	0.999	0.995
1 [Bare Soil]	0.997	0.990	0.996	0.999	0.993
2 [Light Forest]	0.939	0.905	0.937	0.993	0.925
3 [Dense Forest]	0.991	0.995	0.992	0.973	0.993

Diagram 3 Confusion matrix for Mahalanobis classifier (tested on 2017 image).

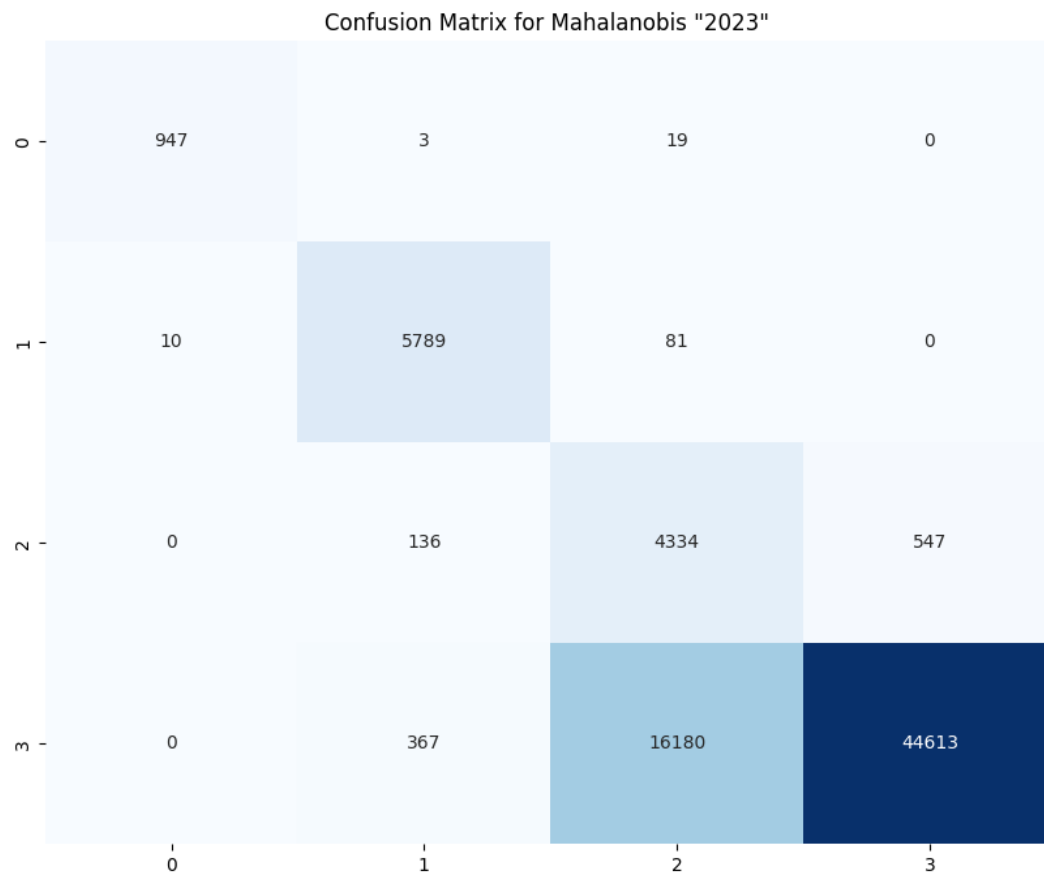


Table 7 Producer's Accuracy Consumer's Accuracy, Sensitivity, Specificity and F1 score for Mahalanobis Classifier

Class	Producer's Accuracy	Consumer's Accuracy	Sensitivity	Specificity	F1 score
0 [Water]	0.979	0.989	0.977	0.999	0.983
1 [Bare Soil]	0.984	0.918	0.984	0.992	0.950
2 [Light Forest]	0.863	0.210	0.863	0.760	0.338
3 [Dense Forest]	0.729	0.987	0.729	0.953	0.839

Diagram 4 Confusion matrix for minimum Distance classifier (tested on 2017 image).

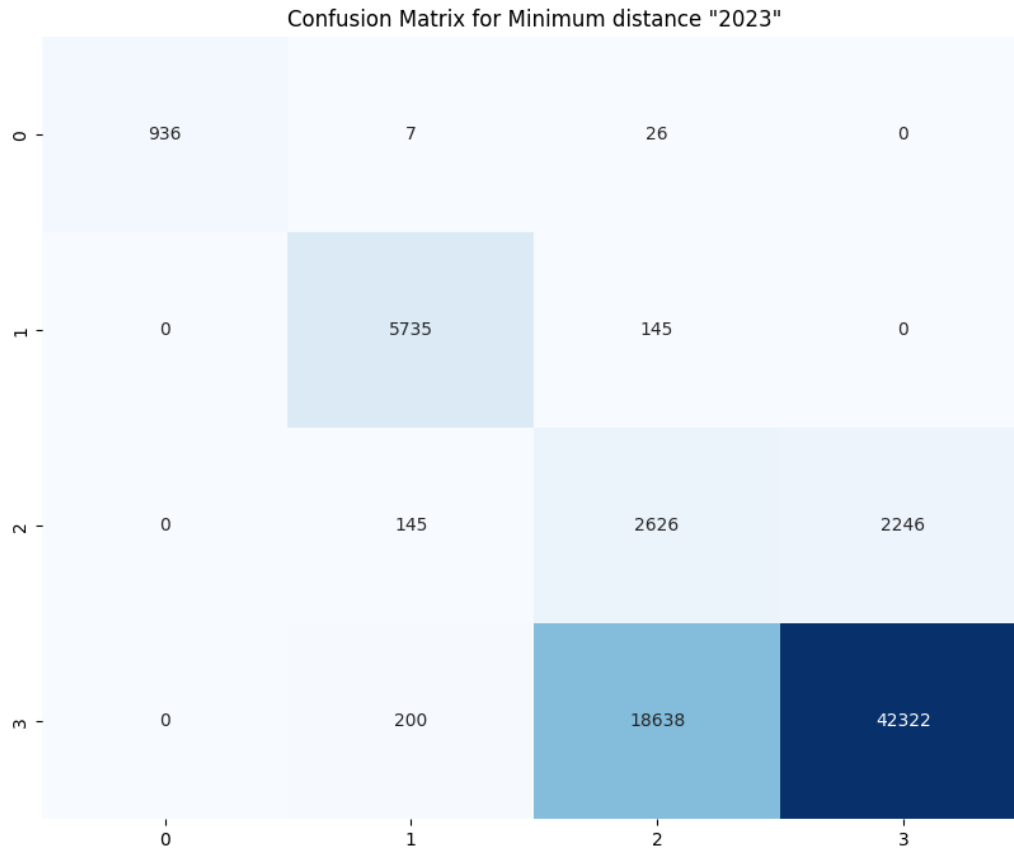


Table 8 Producer's Accuracy Consumer's Accuracy, Sensitivity, Specificity and F1 score for minimum distance Classifier

Class	Producer's Accuracy	Consumer's Accuracy	Sensitivity	Specificity	F1 score
0 [Water]	0.965	1	0.965	1	0.982
1 [Bare Soil]	0.975	0.9421	0.975	0.994	0.958
2 [Light Forest]	0.523	0.122	0.523	0.723	0.198
3 [Dense Forest]	0.692	0.949	0.692	0.810	0.800

Table 9 Overall accuracy and Kappa coefficient for Random Forest Mahalanobis and Minimum distance classifiers.

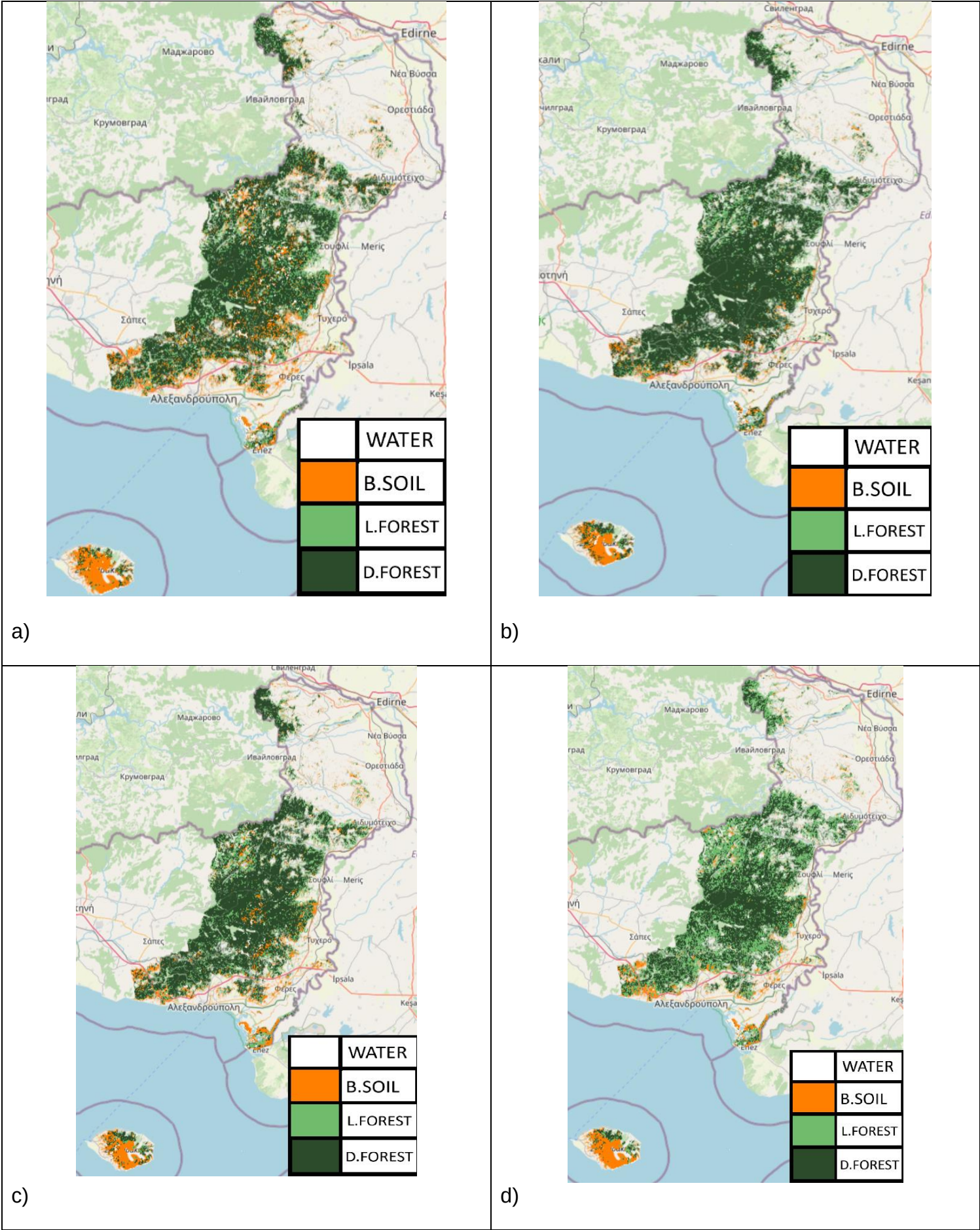
	Overall accuracy	Kappa coefficient
Random Forest	0.988	0.959
Mahalanobis	0.762	0.478
Minimum Distance	0.707	0.365

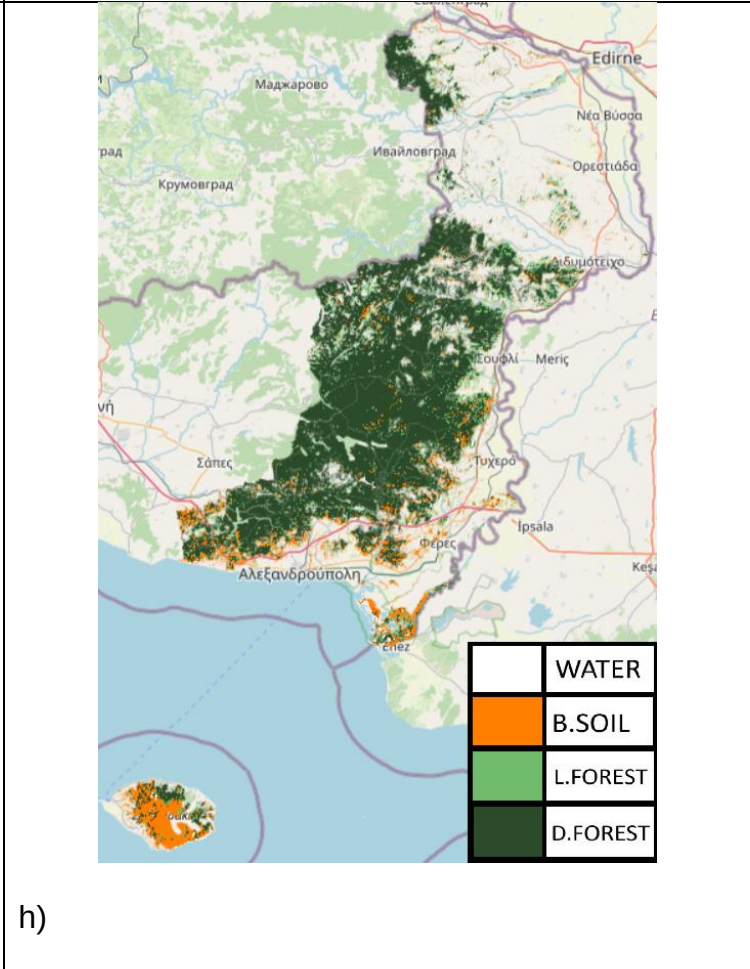
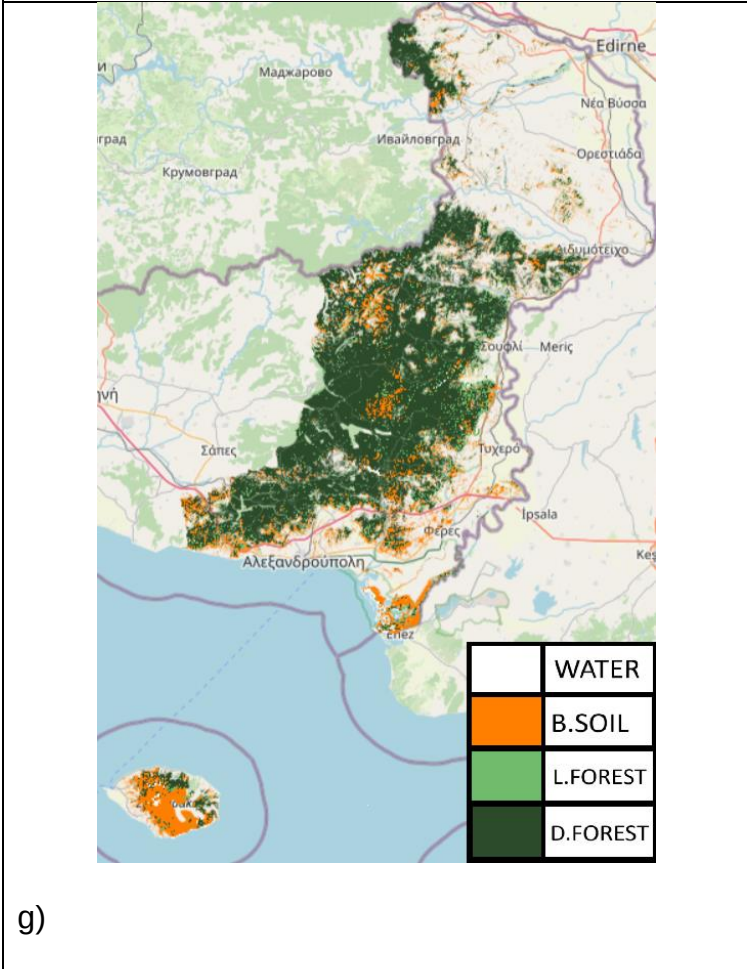
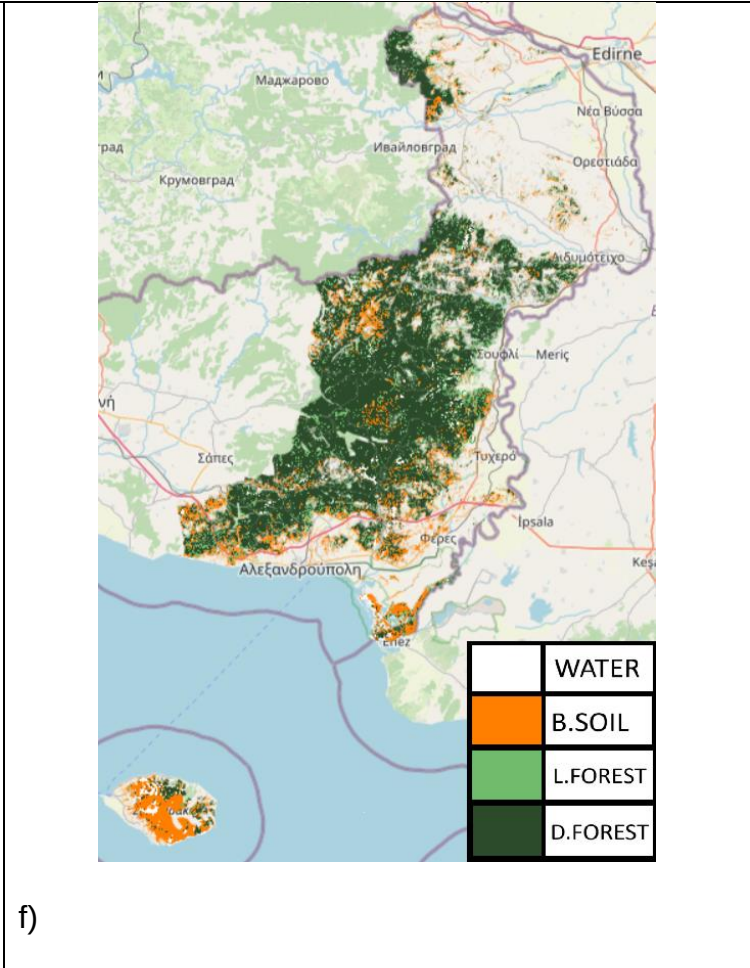
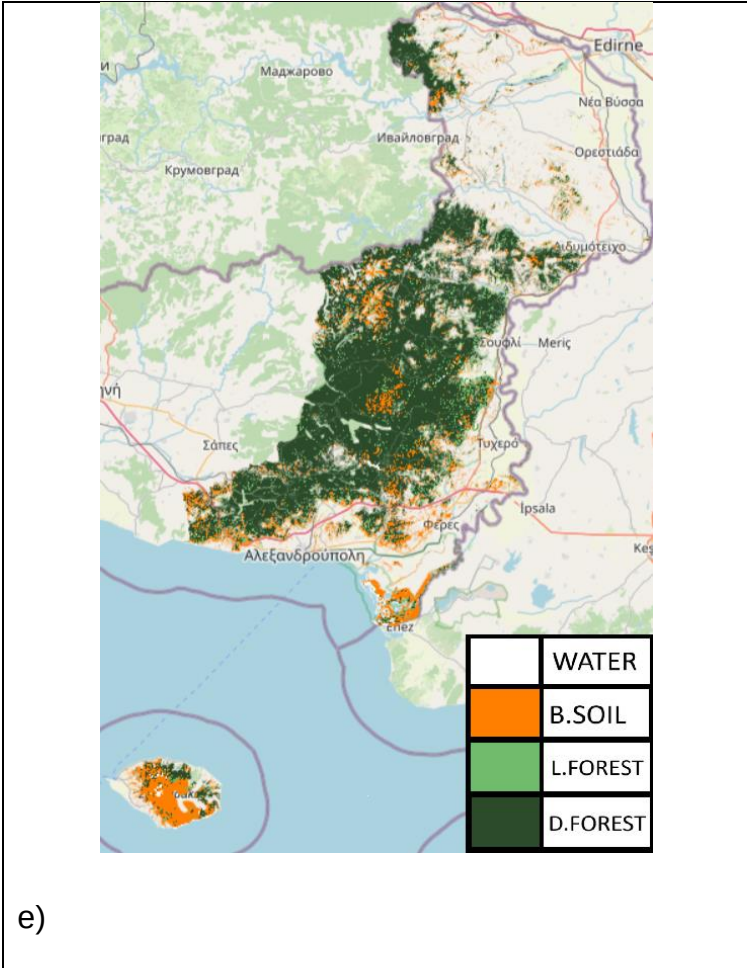
The Random Forest classifier outperforms the other two classifiers when considering key metrics such as the f1 score, sensitivity, and specificity. In direct comparison to the Minimum Distance and Mahalanobis classifiers, the Random Forest classifier, as depicted in **Table 9**, exhibits notably higher performance, with overall accuracy of 0.979 and a Kappa coefficient of 0.937. This becomes evident when contrasted with the corresponding metrics of the other two classifiers with minimum distance having an overall accuracy 0.707 and a Kappa coefficient of 0.365 and Mahalanobis having an overall accuracy 0.762 and a Kappa coefficient of 0.478.

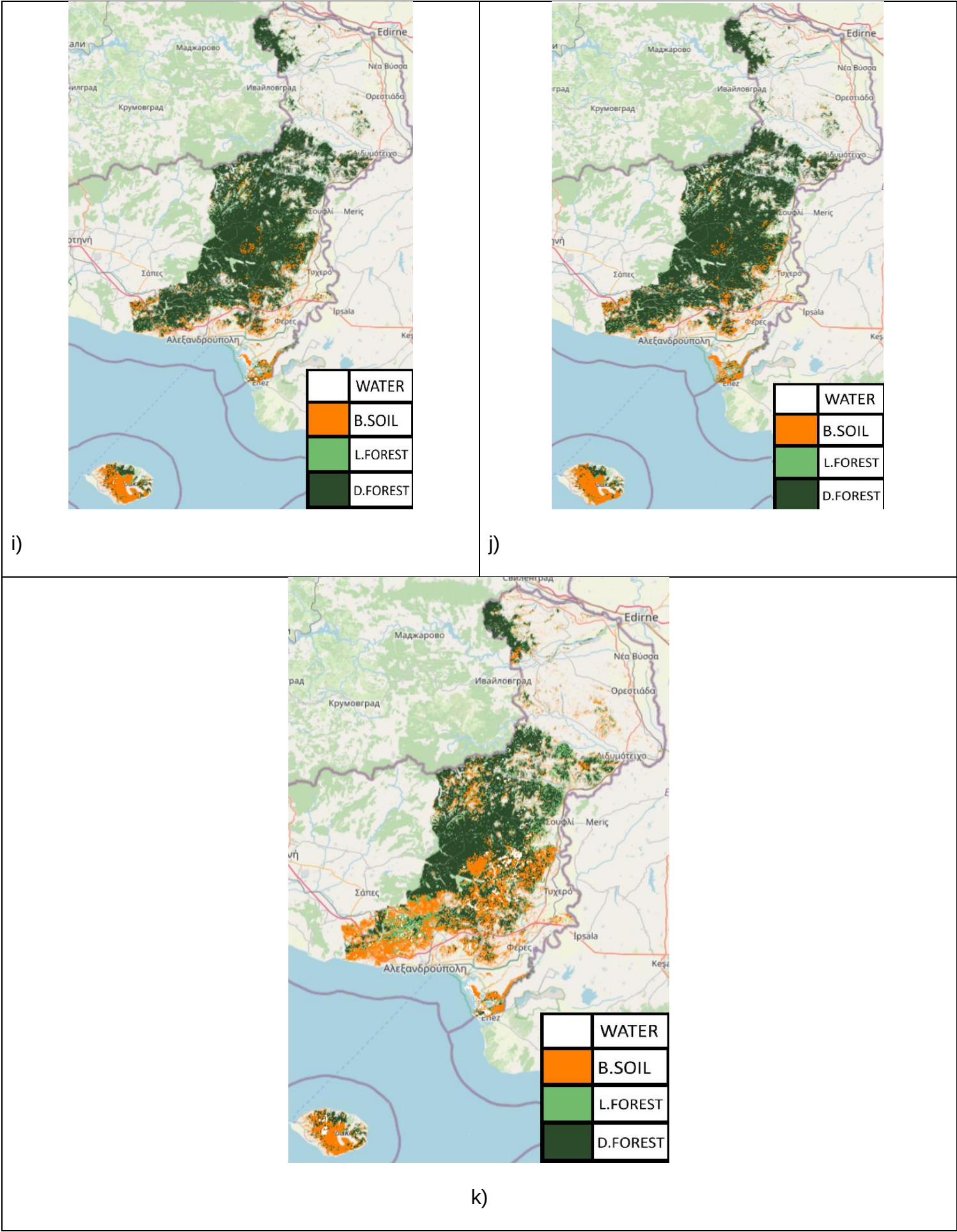
3.1.2 Classification results

After the accuracy assessment for all the classifiers the random forest classifier was applied to the image composites that were created (see Chapter 2.2) and the result of the classifications is shown in **Table 10 (a to k)** below showing the classified images from 2013 to 2023. Although some differences throughout the years are already visible just by observing the classified images, the number of pixels for every class were calculated for every classified image which can be seen in **Diagrams 5,6,7,8,9**. Allowing for a deeper understanding of the different changes that have happened in the forested area over the years.

Table 10 Classified images using Random Forest classifier from 2013 to 2023







In **Diagram 9** the number of pixels classified for every class from 2013 to 2023 separately can be seen. This allows for the changes of every class in comparison to the other classes to be observed. To have a complete image of the changes that happened it is crucial to understand the exact changes for that **Diagrams 5,6,7,8** was created showing all the classes separately making it easier to observe the changes for every class.

Diagram 5 Number of pixels classified for water class every year.

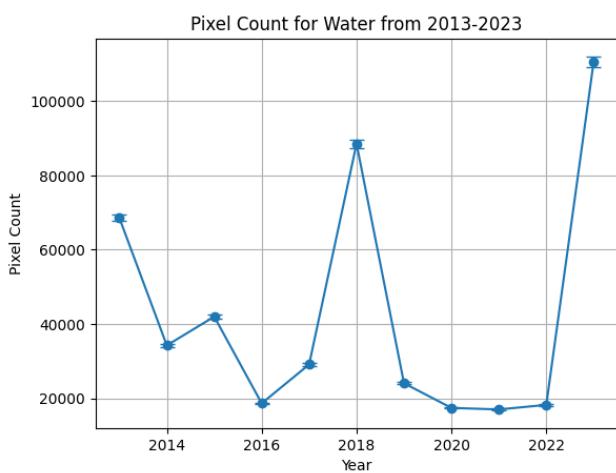


Diagram 6 Number of pixels classified for bare soil class every year.

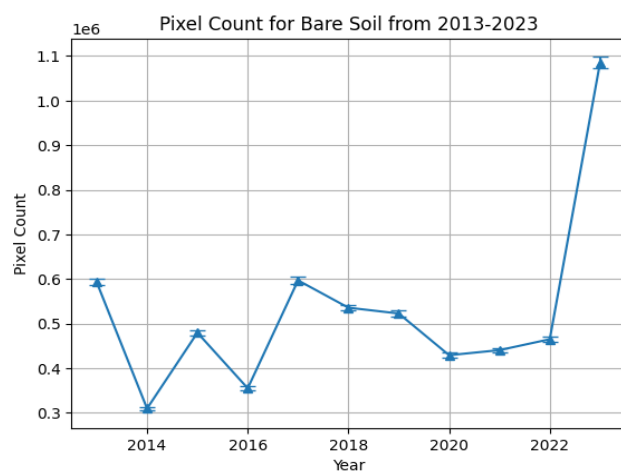


Diagram 7 Number of pixels classified for light forest class every year.

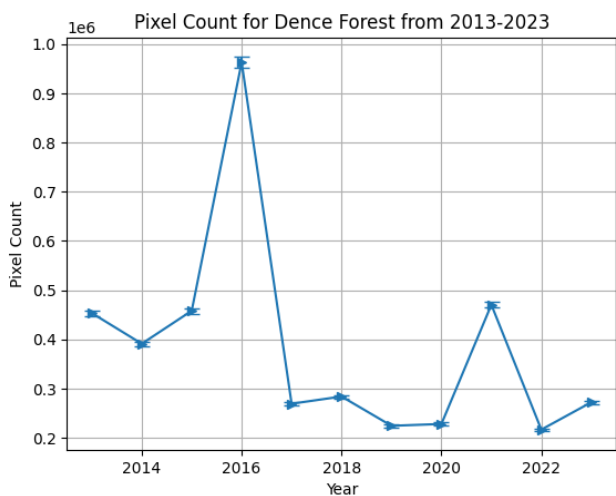


Diagram 8 Number of pixels classified for dense forest class every year.

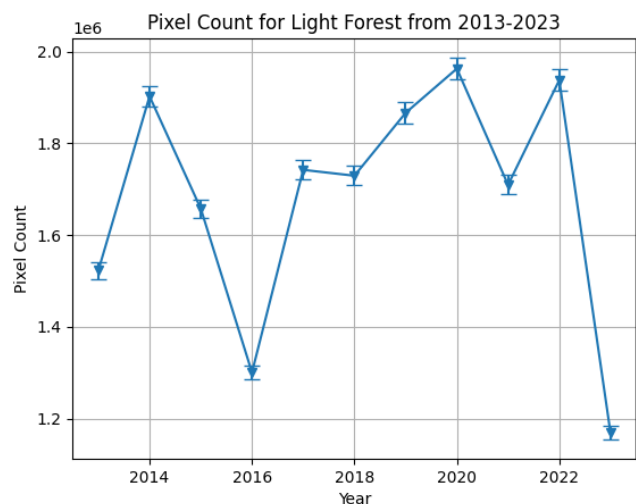
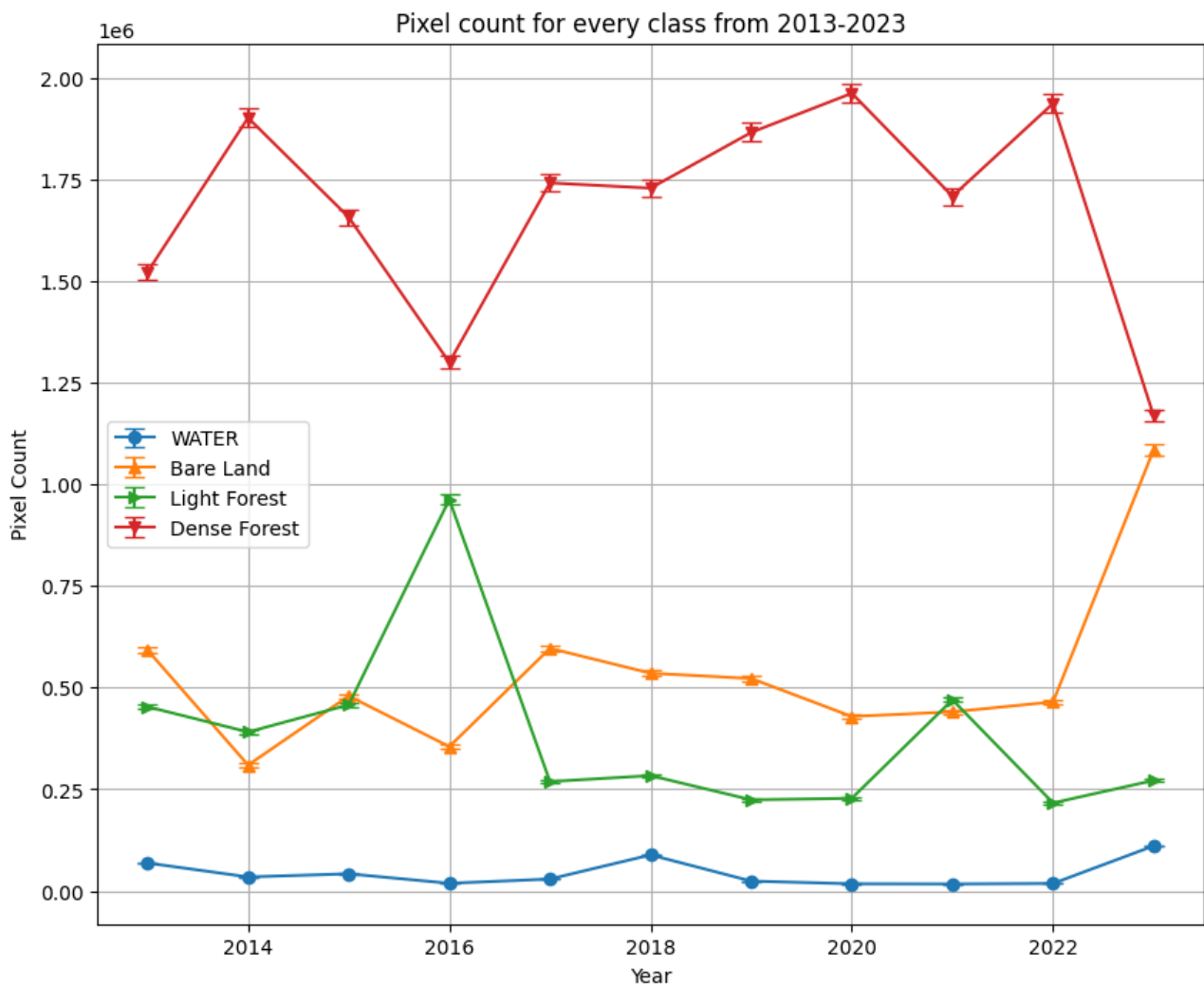


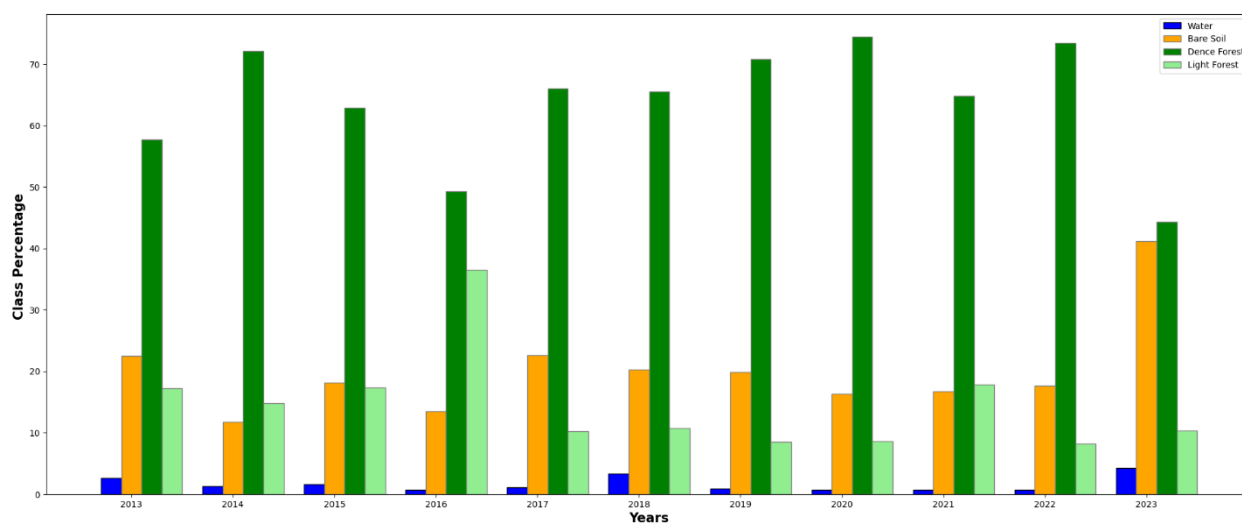
Diagram 9 Pixel count for all classes (Water, bare soil, dense forest, light forest)

A couple of significant changes can be observed in both the graphs and the classified images, the most extreme one being the decrease in the pixels that are classified in the dense forest class in the year 2023 and a corresponding increase in the Bare Land Class . As mentioned in the beginning there was a significant fire in the region of Evros in 2023 that burned 730 hectares of land that was previously mainly forested land making that dramatic decrease of forest land expected in the classification process. A final thing to note is that the increase of the water class although minimal, is because of the smoke due to the fire still being active in that month.

Additionally, noteworthy alterations were observed in the year 2014, where there is a surprising increase in dense forest area in 2014 with a decrease in the bare land followed by a steady decrease of dense forests for the year 2015 and 2016. After that there is a steady increase of dense forest land from 2017 until 2020. In 2020 according to the forest fire maps from the Emergency Response Coordination Center, ERCC [\[18\]](#) there were some

forest fires (fires of a much smaller scale than the fire of 2023) 22 of July of that year in the region which explains the changes.

Diagram 10 Percentage of classified pixels for every class for the years 2013 to 2023.



The same observations can be seen in **Diagram 10** where the percentage of classified pixels is shown corresponding to each year from 2013 to 2023.

Knowing what changes happened during these years is not enough by itself. Where these changes have happened is also very important. To make that possible the classified images of two years are subtracted from each other leaving a number that represents how much the pixel has changed. For example, for a pixel classified in class 2 (Dense forest) in 2013 and a pixel classified in class 0 (Bare Soil) in 2014 in the image 2014-2013 that pixel will have the value of -2 meaning that the pixel has a significant loss. In conclusion negative numbers signify loss of forest land, 0 signifies no change at all and positive numbers signal the development of the forest area. In **Table 11** the color palette and the values that correspond can be observed.

Table 11 Values and color palette for loss and growth.

-2	Significant Loss	
-1	Some Loss	
0	No change	
1	Some Growth	
2	Significant Growth	

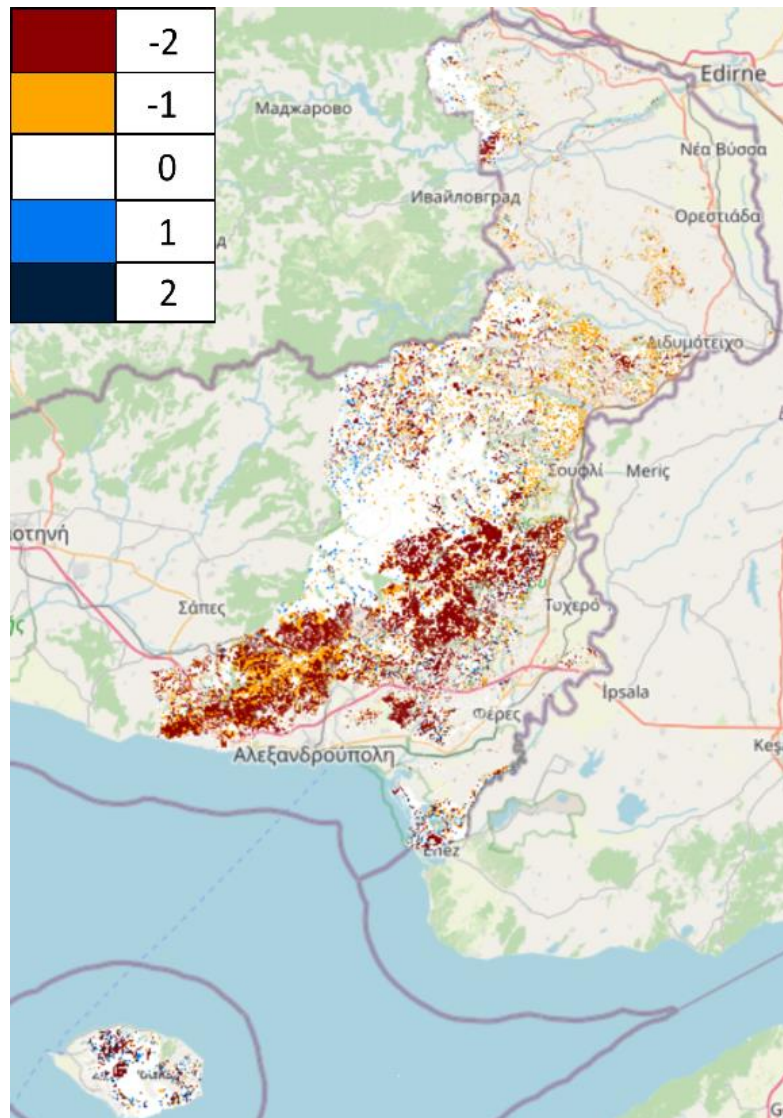


Image 4 : Difference between 2023 – 2022

As mentioned before in the year 2023 a significant increase of bare soil is seen in 2023 alongside with a decrease in densely forested areas. In **Image 4** it is observable that the significant changes were in the north of the region, this is where the fire of 2023 happened in the region of Evros burning a large, forested area.

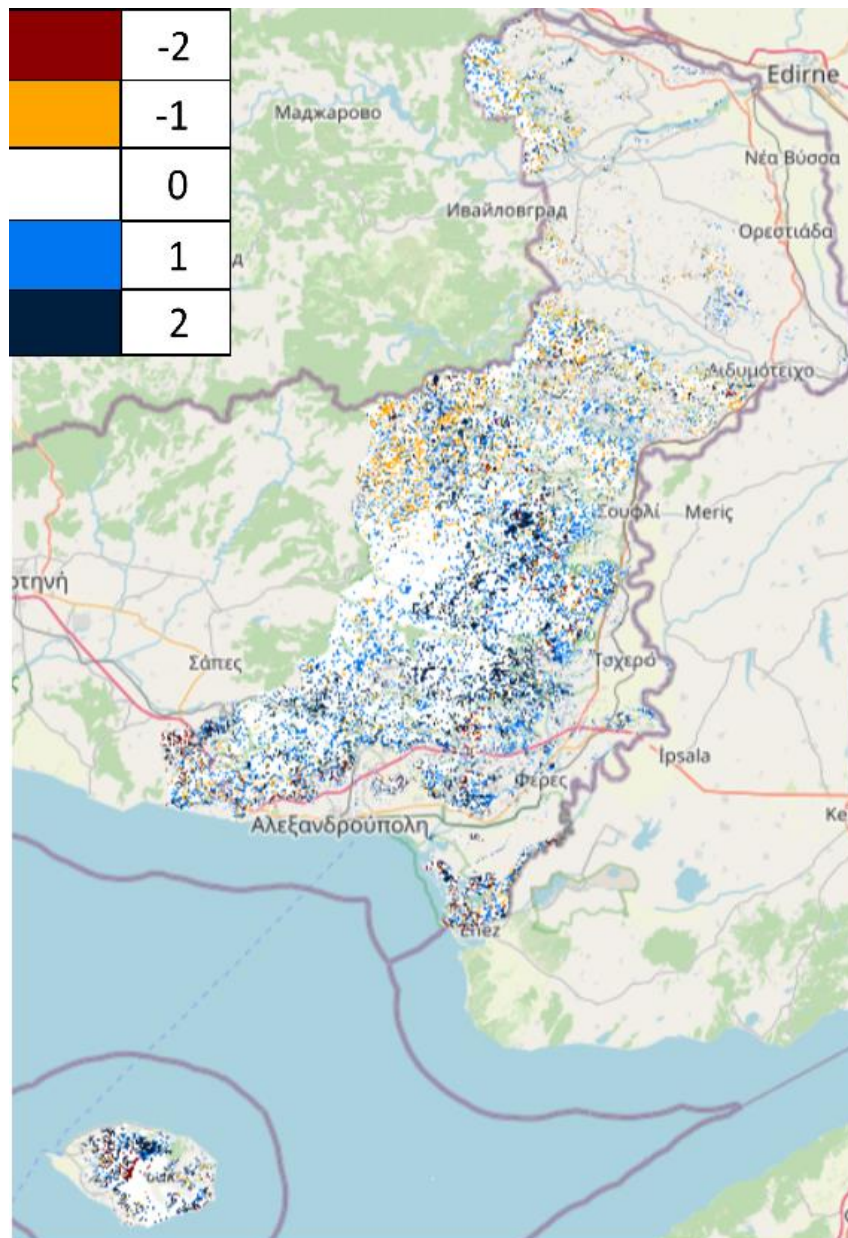


Image 5 Difference between 2014 – 2013

Similarly in **Image 5** the specific areas where the densely forested area has increased can be observed with the value of '2'. There are small areas through the whole Evros region.

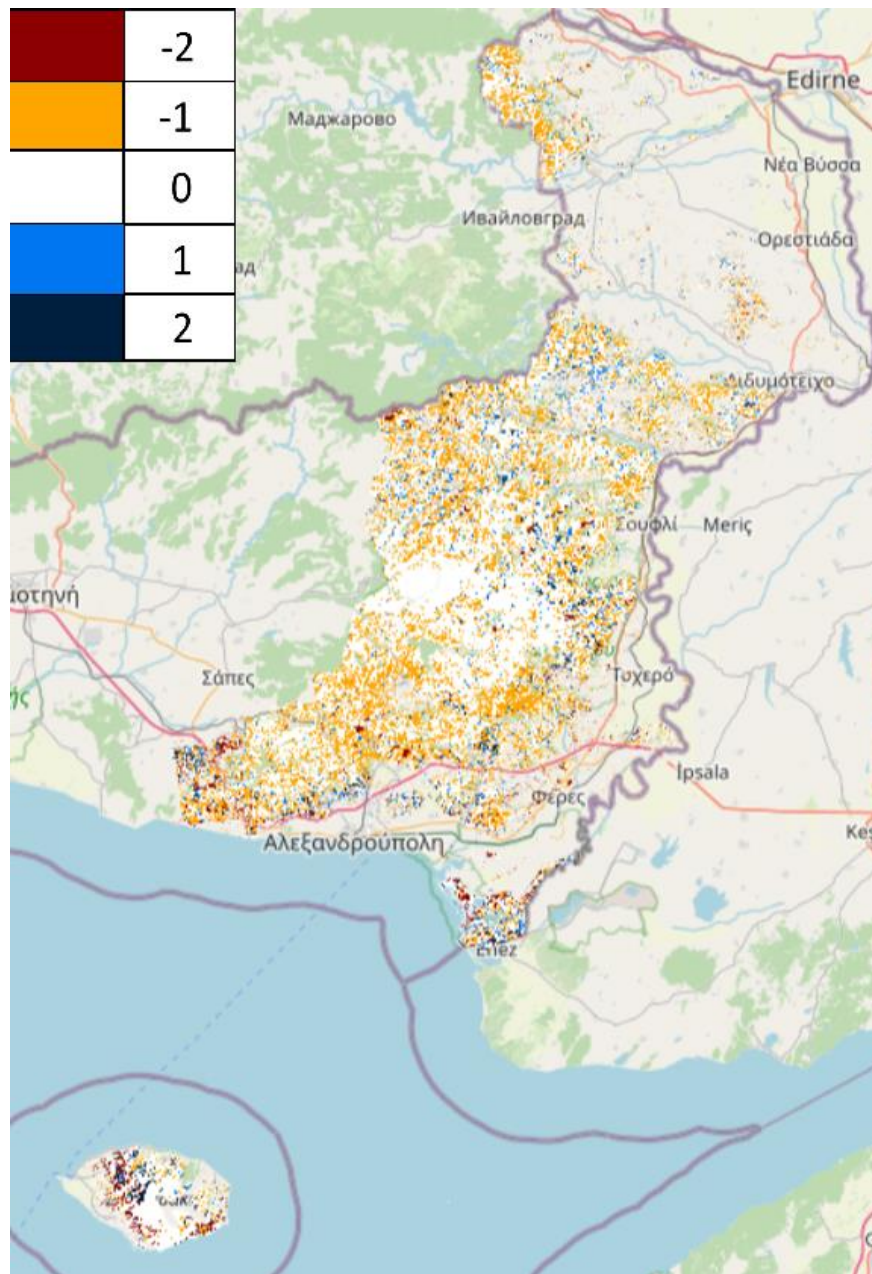


Image 6 Difference between 2016 – 2014

Small steady decrease in dense forests was detected in 2014 to 2016 and from the **Image 6** it is apparent that these changes happened throughout the whole Evros area.

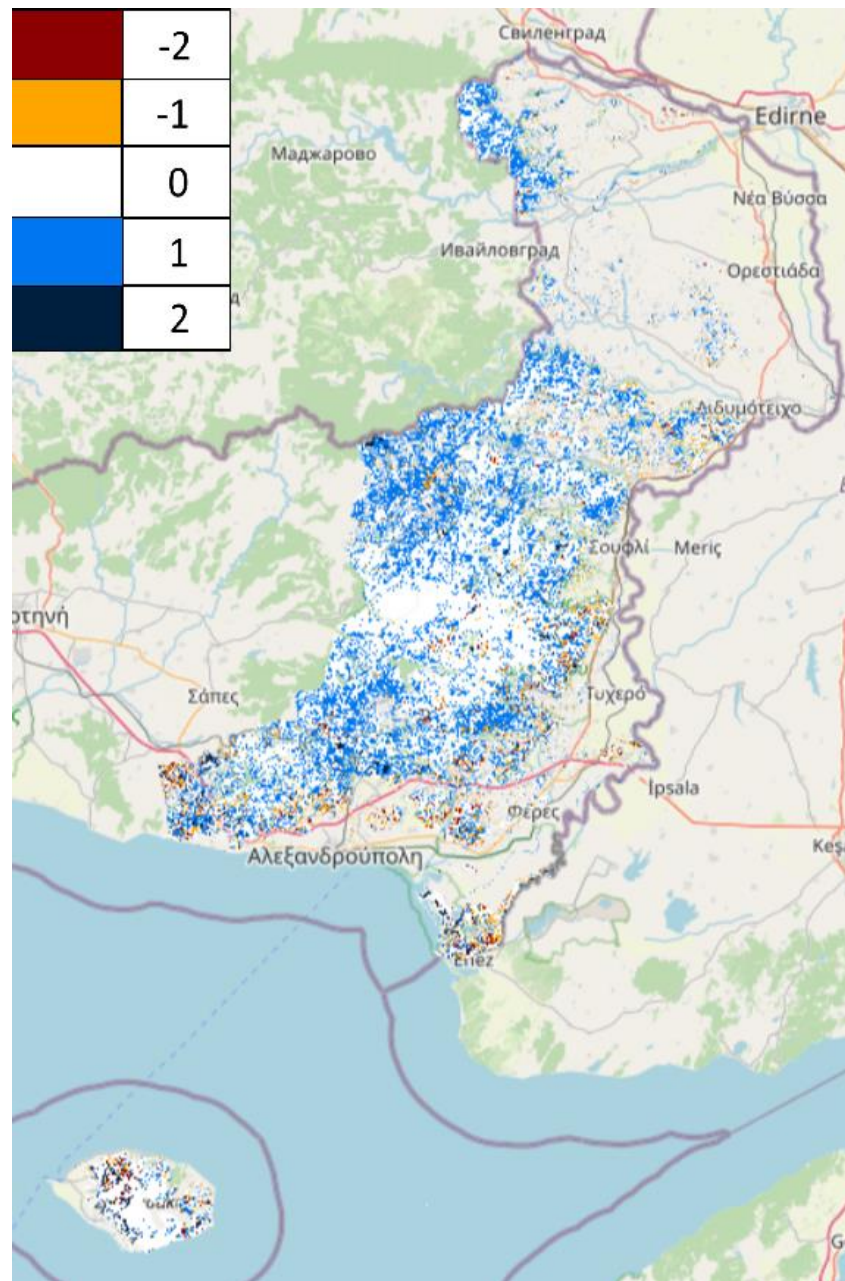


Image7 Difference between 2020 – 2016

Similarly with 2013 -2014 some regrowth can be seen in the years 2016 to 2020 throughout the whole regional unit of Evros shown in **Image 7**.

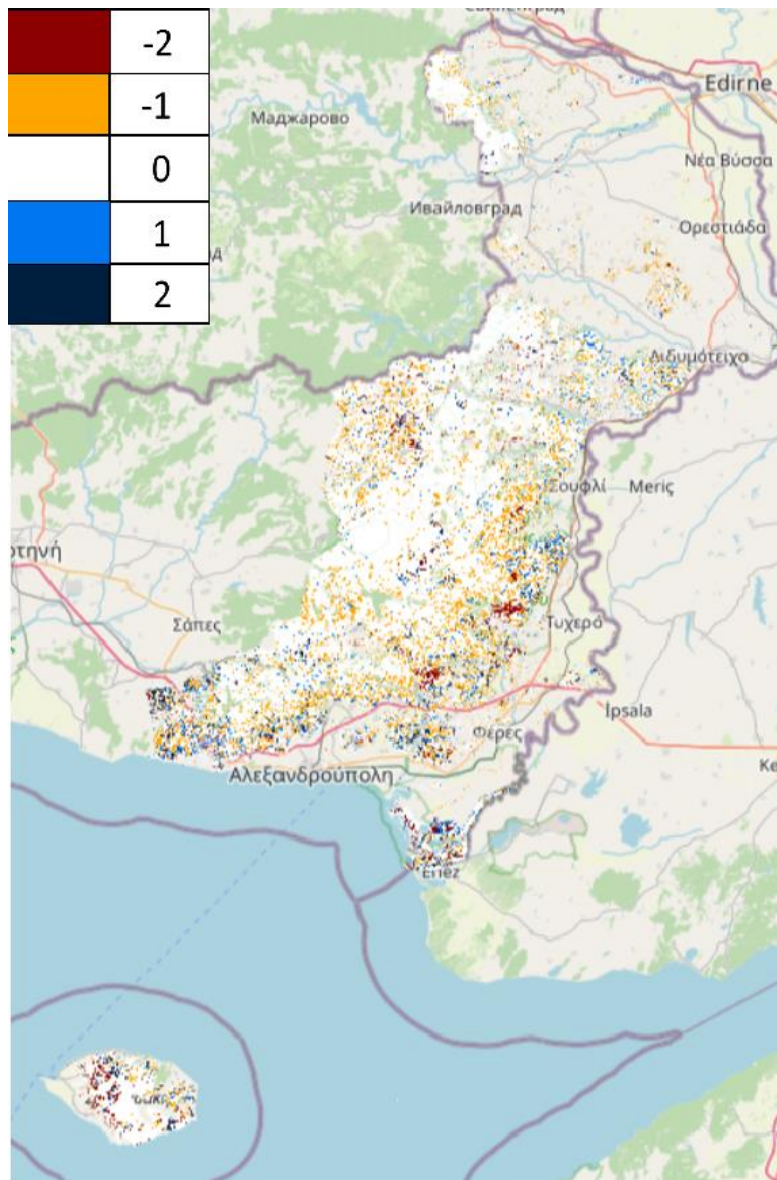


Image 8 Difference between 2021 – 2020

In **Image 8** three very small areas can be seen with values lower than -1. These areas are small fires that happened in July of 2020 [18].

Knowing at this stage where, when and what kind of changes happened in the forested areas of the regional unit of Evros, the areas where the changes happened can be quantified by counting the pixels for every change level from -2 to 2 and multiplying with $900 \cdot 10^{-6}$ shown in **Table 12** .

Table 12 The area and the level that it was changes for the years of interest.

km^2	2023-2022	2014-2013	2016-2014	2020-2016	2021-2020
-2	478.298	35.213	44.809	51.5367	66.1059
-1	330.618	165.190	681.84	142.5537	346.2975
0	1376.777	1518.600	1452.200	1385.0631	1787.418
1	57.699	388.155	136.711	746.9208	111.0168
2	24.328	210.550	34.781	35.4834	53.2296

Using the table above it can be estimated with this analysis that $478.3 [km^2] + 330.618 [km^2] = 808.916 [km^2]$ have been through loss of forested area and $57.699 [km^2] + 24.328 [km^2] = 82.027 [km^2]$ of regrowth between the year 2023 compared with 2022 . In a similar fashion in the years 2020 - 2021 the loss was $411.052 [km^2]$ with $61.052 [km^2]$ of them being significant loss and the remaining $350.000 [km^2]$ being 'some loss', a small amount of regrowth of $168.457 [km^2]$ of which the $53.967 [km^2]$ are areas that have gone through significant loss of forested land. In **Table 13** the sum of the area loss and growth can be seen for every timespan of interest that was previously mentioned.

Table 13 Complete Forest area loss and growth for the years of interest.

[km^2]	2023-22	2014-13	20216-14	2016-20	2021-20
LOSS	808.916	200.403	726.649	194.0090	412.403
GROWTH	82.027	598.705	171.492	782.404	164.246

4. DISCUSSION

The protection of forested areas is very important, especially in the last years due to the significant increase of wildfires along with the yearly rising temperatures worldwide. One crucial step that can be very useful in the instance of a wildfire is the estimation of the loss of the forested area. Calculating loss of forested land can be very useful not only in instances of wildfires but also to monitor changes in a forest during a certain timespan, this can assist in understanding smaller changes and trends. Studying a forested area can also help in understanding how the area "responds" in the rising temperatures of the planet, acquiring

information from that can be a crucial resource helping policymakers and relevant authorities to choose the next steps for the protection of the area.

After collecting the composite images using google earth engine and creating the samples for the training dataset, after an accuracy assessment of the three classifiers that were implemented was calculated, Random Forest classifier with four branches, Minimum distance classifier and Mahalanobis classifier, the one that works best with this dataset is the Random Forest classifier with four branches with an overall accuracy of 0.988 and kappa coefficient 0.959. After that by calculating the pixels that are classified on every class for the classes water, bare soil, light forest, dense forest, years that have severe changes were observed and then mapped in order to locate where exactly these changes have happened.

4.1 Overview of findings

The most significant changes that were observed are in the years 2020-2021 with 411.052 [km^2] of lost forest area in the south east of the regional unit, caused by three relatively small forest fires and 2022-2023 with 808.916 [km^2] of lost area due to forest fires located in the south of the regional unit of Evros, a very significant forest fire in 2023. Some other noteworthy changes are in the span of 2014 to 2016 there where 726.649 [km^2] of land had loss of forested area from which 681.4 of them were of some loss. In all the aforementioned years there was some regrowth of forested land but the only year there was more regrowth than loss was 2013-2014 with an overall growth of 598.705 [km^2] with 388.155 of them being some growth.

4.2 Comparison with Existing Literature

According to a publication from the Department of Geography, Harokopio University of Athens, where Copernicus Land Service and Sentinel-2 Satellite Mission Data were used to Assess the Spatial Dynamics and Distribution of the Extreme Forest Fires of 2023 in Greece [21] one of the fires that were analyzed is the Evros fire of 2023 calculating a burnt area of 903.04 [km^2], in this analysis the area that was calculated to be lost in 2023 is 808.916 [km^2]. The difference between this analysis and the aforementioned publication [21] is because this analysis does not take into account agricultural land and civilizations in the area. It is also worth noting that the Sentinel 2 data in the publication [21] have a spatial analysis of 10 metres, in comparison to the 30 metres of the Landsat 8 image dataset that

was used in this analysis causing some loss of information due to the bigger area represented in each pixel.

4.3 Limitations

The main limitations of this thesis are the spatial resolution of the satellite images used are only thirty (30) meters making the distinction of forested land possible but making the distinction of different types of forests difficult although in this area the forested land is mostly uniform as to its forest type mainly being Thermophilus deciduous type.

5. CONCLUSION

Forests play a crucial role for life on earth, maintaining sustainable conditions for the living organisms that inhabit it like air quality and serving as a habitat for earth's fauna and flora to coexist and evolve together with the passing of time. However, the world's forests are facing unprecedented challenges due to deforestation, climate change, and human encroachment. These threats not only endanger the delicate balance of ecosystems but also jeopardize the essential services forests provide to humanity.

Governments and citizens alike are increasingly recognizing the urgency of preserving forests, understanding their pivotal role in maintaining a healthy environment for humanity. Recent events, such as the devastating wildfires that swept through Greece in 2023, serve as stark reminders of the fragility of these ecosystems and the pressing need for concerted conservation efforts. The widespread destruction caused by such disasters underscores the importance of proactive measures to safeguard forests against the escalating threats of climate change and human activities. In the current age it is very important for governments and their citizens to protect and help the preservation of forested areas because they play a crucial role fostering a healthier environment for humanity both mentally and physically making it an asset that needs to be protected from the rising temperatures that have past and are to be repeated in later years. In the country of Greece there were a lot of wild forest fires in the year of 2023 specifically. The forest wildfire of the Evros was one of the biggest ones, having a significant amount of forested area burned and lost according to a study from the Department of Geography, Harokopio University of Athens and Centre for Research & Technology Hellas (CERTH) [21], 912.04 km² of area both forested and agricultural land was burned.

A very important resource for scientists in times like this is data and one source of very versatile and abundant as mentioned before is satellite data with the range of spectral information that can be found and the historical archive that has been built over time and are freely available online via multiple platforms.

In this thesis a ten (10) year forest change analysis of the protected forested areas in the region of Evros was done using the Google Earth Engine tools in order to get access to both the data from Landsat 8 satellites and for the classification and analysis of the data . The data are accessed by the Google Earth Engines API allowing for easy access and usage of satellite data as well as cloud computing helping users to not depend on personal resources except an internet connection along with Google Collaboratory in order for the code to be easily accessible and take advantage of cloud computing also. The Google Earth Engine API was used to access Landsat 8 raw images for the region of Evros from 2013 to 2023 , calculate NDVI (Normalized Difference Vegetation Index) and BSI (Bare Soil Index) for every image , train the Mahalanobis , random forest and minimum distance classifiers with polygons that have been extracted via google Earth engine Code editor , accuracy assessment process by calculating the kappa coefficient , overall accuracy etc in order to make the best choice for the classifier that was used in this case being the random forest classifier and finally calculating the changes in the forested area of the region via comparison of the classified images.

After the completion of the analysis the outcome is that in the year 2023 there was a loss of forested aerie in the region of 808.916 km², 478.298 of them being of significant loss of forested aerie and the rest (330.618) being of some loss. Some loss that seems to be from a fire specifically 3 small fires in the region the year 2020. And finally, some slight decrease in the forested area can be observed between the years 2014 to 2016 that can be because of changes in the weather or other external factors. After that from 2016 to 2020 there was a consistent small and steady increase of forested area every year.

It is important to note that this process can be recreated again for future analysis (adding for example 2024 25 etc) and can be shared via google Collaboratory. In works like this it is imperative to consider future steps in order to build and improve upon the original idea. Future direction based on this thesis can be the prediction of regrowth in forested areas that have been burned based on growth in the region in previous years using machine learning algorithms. A bigger time series can be used for example from 1990 to 2024 by combining Landsat time series or using different satellite data.

ABBREVIATIONS – ACRONYMS

GEE	Google Earth Engine
NDVI	Normalized Difference Vegetation Index
BSI	Bare Soil Index
OLI	Operational Land Imager
TIRS	Thermal Infrared Sensor
API	Application Programming Interface

ANNEX I

This annex provides links to the GitHub repository and Google Collab notebook that contain the source code referenced in this report. These resources serve as supplementary materials for further exploration and verification of the findings presented <https://github.com/stellamala/Forest-change-analysis-in-the-Evros-region->. The Google Collaboratory notebook can be accessed at the following link : https://colab.research.google.com/drive/1YKdQMyi6P07KIdJQbzGTi_P6fD6iRkAW?usp=drive_link

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